

Stationary source emissions — Determination of mass concentration of sulphur dioxide — Reference method

The European Standard EN 14791:2005 has the status of a
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National foreword

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Emissions de sources fixes - Détermination de la concentration massique du dioxyde de soufre - Méthode de référence

Emissionen aus stationären Quellen - Bestimmung der Massenkonzentration von Schwefeldioxid - Referenzverfahren

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Foreword

This European Standard (EN 14791:2005) has been prepared by Technical Committee CEN/TC 264 "Air quality", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by May 2006, and conflicting national standards shall be withdrawn at the latest by May 2006.

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For relationship with EU Directive(s), see informative Annex ZA, which is an integral part of this European Standard.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

1 Scope

This European Standard describes a manual method for sampling and determining SO₂ content in ducts and stacks emitting to the atmosphere by two analytical methods: Ion chromatography and Thorin method.

This European Standard is the Standard Reference Method (SRM) for periodic monitoring and for calibration or control of Automatic Measuring Systems (AMS) permanently installed on a stack, for regulatory purposes or other purposes. To be used as the SRM, the user shall demonstrate that the performance characteristics of the method are better than the performance criteria defined in this European Standard and that the overall uncertainty of the method is less than $\pm 20,0$ % relative at the daily Emission Limit Value (ELV).

An Alternative Method to this SRM may be used provided that the user can demonstrate equivalence according to the Technical Specification CEN TS 14793, to the satisfaction of his national accreditation body or law.

This Standard Reference Method has been evaluated during field tests on waste incineration, co-incineration and large combustion installations. It has been validated for sampling periods of 30 min in the range of (0,5 to 2 000) mg/m³ SO₂ for Ion Chromatography variant and 5 mg/m³ to 2 000 mg/m³ SO₂ for Thorin method according to emission limit values laid down in the following Council Directives:

- Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants;
- Council Directive 2000/76/EC on waste incineration plants.

The limit values of EU Directives are expressed in mg SO₂/m³, on dry basis and at the reference conditions of 273 K and 101,3 kPa.

2 Normative references

The following referenced documents are indispensable for the application of this European Standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ENV 13005, *Guide to the expression of uncertainty in measurement*.

EN 13284-1, *Stationary source emissions — Determination of low range mass concentration of dust — Part 1: Manual gravimetric method*.

CEN/TS 14793, *Stationary source emission — Intralaboratory validation procedure for an alternative method compared to a reference method*.

3 Terms, definitions, symbols and abbreviations

3.1 Terms and definitions

For the purposes of this European Standard, the following terms and definitions apply.

3.1.1

absorber

device in which sulphur oxide is absorbed into an absorption liquid

3.1.2**absorption efficiency (ε)**

ratio ε of quantity of the analyte q_1 collected in the first absorber divided by the quantity of the analyte collected in the first and the second absorber ($q_1 + q_2$)

$$\varepsilon = q_1 / (q_1 + q_2)$$

3.1.3**analytical detection limit (L_D)**

concentration value of the measurand below which there is at least 95 % level of confidence that the measured value corresponds to a sample free of that measurand

3.1.4**automatic measuring system (AMS)**

measuring system permanently installed on site for continuous monitoring of emissions

NOTE 1 An AM is a method which is traceable to a reference method.

NOTE 2 Apart from the analyser, an AMS includes facilities for taking samples (e.g. probe, sample gas lines, flow meters, regulators, delivery pumps) and for sample conditioning (e.g. dust filter, moisture removal devices, converters, diluters). This definition also includes testing and adjusting devices, that are required for regular functional checks.

3.1.5**calibration of an AMS**

statistical relationship between values of the measurand indicated by the measuring system (AMS) and the corresponding values given by the standard reference method (SRM) used during the same period of time and giving a representative measurement on the same sampling plane

NOTE The result of calibration permits to establish the relationship between the values of the SRM and the AMS (calibration function).

3.1.6**calibration of the SRM**

set of operations that establish, under specified conditions, the relationship between values of quantities indicated by a measuring instrument or measuring system, and the corresponding values realized by standard reference methods implemented in the state of the art in order to provide representative results

[VIM]

3.1.7**chemical blank value**

sulphate ion content of an unexposed sample of the absorption solution, plus reagents that are added to the solution before analysis if necessary

3.1.8**emission limit value (ELV)**

emission limit value according to EU Directives on the basis of 30 min, 1 h or 1 day

3.1.9**field blank**

value determined by a specific procedure used to ensure that no significant contamination has occurred during all steps of the measurement and to check that the operator can achieve a quantification level adapted to the task

3.1.10

influence quantity

quantity that is not the measurand but that affects the result of the measurement

[adapted VIM 2.7]

Examples

- ambient temperature;
- atmospheric pressure;
- presence of interfering gases in the flue gas matrix.

3.1.11

measurand

particular quantity subject to measurement

[VIM 2.6]

3.1.12

measurement series

several successive measurements carried out on the same sampling plane and at the same process operating conditions

[EN 13284-1]

3.1.13

measuring system

complete set of measuring instruments and other equipment assembled to carry out specified measurements

[VIM 4.5]

3.1.14

performance characteristic

one of the quantities (described by values, tolerances, range, ...) assigned to equipment in order to define its performance

3.1.15**repeatability in the laboratory**

closeness of the agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement

NOTE 1 Repeatability conditions include:

- same measurement procedure;
- same laboratory;
- same sampling equipment, used under the same conditions;
- same location;
- repetition over a short period of time.

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the repeatability is expressed as a value with a level of confidence of 95 %.

[VIM 3.6]

3.1.16**repeatability in the field**

closeness of the agreement between the results of simultaneous measurements of the same measurand carried out with two equipments under the same conditions of measurement

NOTE 1 These conditions include:

- same measurement procedure;
- two equipments, the performances of which are fulfilling the requirements of the reference method, used under the same conditions;
- same location;
- implemented by the same laboratory;
- typically calculated on short periods of time in order to avoid the effect of changes of influence parameters (e.g. 30 min).

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the repeatability under field conditions is expressed as a value with a level of confidence of 95 %.

3.1.17

reproducibility in the field

closeness of the agreement between the results of simultaneous measurements of the same measurand carried out with several sets of equipment under the same conditions of measurement

NOTE 1 These conditions are called field reproducibility conditions and include:

- same measurement procedure;
- several sets of equipment, the performance of which fulfils the requirements of the reference method, used under the same conditions;
- same location;
- implemented by several laboratories.

NOTE 2 Reproducibility may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the reproducibility under field conditions is expressed as a value with a level of confidence of 95 %.

3.1.18

sampling location

specific area close to the sampling plane where the measurement devices are set up

3.1.19

sampling plane

plane normal to the centreline of the duct at the sampling position

[EN 13284-1]

3.1.20

sampling point

specific position on a sampling plane at which a sample is extracted

[EN 13284-1]

3.1.21

standard reference method (SRM)

measurement method recognised by experts and taken as a reference by convention, which gives, or is presumed to give, the accepted reference value of the concentration of the measurand (3.1.11) to be measured

3.1.22

uncertainty

parameter associated with the result of a measurement, that characterises the dispersion of the values that could reasonably be attributed to the measurand

3.1.22.1

standard uncertainty u

uncertainty of the result of a measurement expressed as a standard deviation u

3.1.22.2

combined uncertainty u_c

standard uncertainty u_c attached to the measurement result calculated by combination of several standard uncertainties according to GUM

3.1.22.3**expanded uncertainty U**

quantity defining a level of confidence about the result of a measurement that may be expected to encompass a specific fraction of the distribution of values that could reasonably be attributed to a measurand

$$U = k \times u$$

NOTE In this European Standard, the expanded uncertainty is calculated with a coverage factor of $k = 2$, and with a level of confidence of 95 %.

3.1.22.4**overall uncertainty U_c**

expanded combined standard uncertainty attached to the measurement result calculated according to GUM

$$U_c = k \times u_c$$

3.1.23**uncertainty budget**

calculation table combining all the sources of uncertainty according to EN ISO 14956 or ENV 13005 in order to calculate the overall uncertainty of the method at a specified value

3.2 Symbols

C_m	mass concentration of sulphur dioxide in the sample gas, in milligrams per cubic metre (of gas)
Cl_r	repeatability confidence interval, in milligrams per cubic metre
Cl_R	reproducibility confidence interval, in milligrams per cubic metre
f_a	equivalent mass of sulphur dioxide of 1 mm of titration solution (barium perchlorate standard volumetric solution) used for titration in Thorin method, in milligrams per millilitre
f_v	ratio of the volume of the pre-treated sample solution (sample absorption solution pre-treated before analyse) to the volume of the aliquot taken for the titration in Thorin method
L_D	limit of detection, in milligrams per litre of SO_4^{2-}
L_Q	limit of quantification, in milligrams per litre of SO_4^{2-}
m_s	weight of the sample solution (absorption solution used for sampling + rinsing solution), in grams
P_m	absolute pressure at the gas volume meter, in kilopascals
P_{std}	standard pressure: 101,3 kPa
$P_{sat}(T_m)$	saturation vapour pressure of water at volume gas meter temperature, in kilopascals
q_s	mass concentration of sulphate in sample absorption solution, in milligrams per litre (of solution)
q_{cb}	mass concentration of sulphate in chemical blank solution, in milligrams per litre (of solution)
r	repeatability, in milligrams per cubic metre or percentage
R	reproducibility, in milligrams per cubic metre or percentage

R_s	peak resolution S_s : volume of titration solution used for titration of sample absorption solution, in millilitre
S_{cb}	volume of titration solution used for titration of chemical blank solution, in millilitre
S_r	repeatability standard deviation, in milligrams per cubic metre or percentage
S_R	reproducibility standard deviation, in milligrams per cubic metre or percentage
$S_{r\ limit}$	maximum allowable repeatability standard deviation, in milligrams per cubic metre
S_s	volume of titration solution used for the titration of the aliquot of the pre-treated sample solution, in millilitres
t_1	retention time of the first peak, in seconds
t_2	retention time of the second peak, in seconds
T_j	temperature at the gas meter, in Kelvin
T_m	mean temperature at the gas volume meter in Kelvin
T_{std}	standard temperature, 273 K
V_1	reading at the gas volume meter at the beginning of the sampling period in cubic metres
V_2	reading at the gas volume meter at the end of the sampling period in cubic metres
$V_{m\ (std)}$	dry gas volume measured, corrected to standard conditions, in cubic metres
V_s	volume of the sample solution (absorption solution used for sampling + rinsing solution), in litres
w_1	peak width, on the time axis, of the first peak, in seconds
w_2	peak width on the time axis, of the second peak, in seconds
ε	absorption efficiency in percentage
σ	conductivity in micro-siemens per metre
ρ_{20}^4	density of a liquid at 20 °C compared to water's at 4 °C in kilogrammes per litre
χ	volume content in percentage

3.3 Abbreviations

PE	Polyethylene
PTFE	Polytetrafluoroethene

4 Principle

4.1 General

This European Standard describes the Standard Reference Method (SRM) with two alternative analytical techniques for determining sulphur dioxide (SO_2) content emitting to atmosphere from ducts and stacks. The specific components and the requirements for the measuring system are described. A number of performance characteristics with associated minimum performance criteria are specified for the measuring system (see Tables 2 and 3 in 10.2). These performance characteristics and the overall uncertainty of the method shall meet the specifications given in this European Standard.

4.2 Measuring principle

A representative sample of gas is extracted via a heated temperature-controlled probe. The sample is filtered and drawn through hydrogen peroxide absorber solutions for a specified time and at a controlled flow rate. The sulphur dioxide in the sampled gas is absorbed and oxidised to sulphate ion. The mass concentration of sulphate in the absorption solutions is subsequently determined using ion chromatography or by titration with a barium perchlorate solution using Thorin as indicator. SO_3 is also absorbed and transformed in sulphate ion and is therefore an interferent.

This European Standard has been validated with a 0,3 % H_2O_2 absorption solution up to 1 000 mg/m^3 and with a 3,0 % H_2O_2 absorption solution up to 2 000 mg/m^3 . Typical concentration of the absorption solution is 0,3 % H_2O_2 . However, for concentrations higher than 1 000 mg/m^3 it is recommended, in case of a bad efficiency either to decrease the flow or to increase the concentration.

5 Reagents

5.1 General

During the analysis, use only reagents of recognised analytical grade.

Normal, accepted laboratory safety practices and cleaning procedures for glassware should be followed during reagent preparation.

WARNING — Use the reagents in accordance with the appropriate health and safety regulations.

5.2 Hydrogen peroxide

Commercially available solution of H_2O_2 ; mass content 30 %, $\rho_{20}^4 = 1,11 \text{ kg}/\text{l}$.

5.3 Water

H_2O ; ultra pure water with conductivity $\sigma < 10 \mu\text{S m}^{-1}$.

5.4 Absorption solution, H_2O_2

The absorption solution is a hydrogen peroxide solution (5.2) diluted to a mass concentration of 0,3 % H_2O_2 in water (5.3). For concentrations higher than 1 000 mg/m^3 it is suggested in case of a bad efficiency, either to decrease the flow, or increase the concentration of the absorption solution.

For the preparation of the mass concentration of 0,3 % H_2O_2 in water, thoroughly mix about 10 ml of 30 % of H_2O_2 (5.2) with 500 ml of water (5.3) and make up to 1 000 ml with water (5.3). Store the solution in a glass or PE bottle in a dark place and for no longer than one week.

WARNING — Decomposition of the solution may occur and may lead to the explosion of the storage bottle. It is recommended that the lid of this bottle is not closed too tightly or to use a security cap.

NOTE Cleanliness of the glassware is important to avoid a possible decomposition of hydrogen peroxide.

5.5 Reagents for chromatographic analysis

5.5.1 Eluent solution

The choice of eluent depends on the manufacturer's separator column and detector. For the exact composition of the eluent, use a validated solvent for the method and/or refer to the instructions given by the manufacturer.

NOTE For an ion chromatograph using the suppressor technique, a typical eluent is a solution of $1,7 \times 10^{-3}$ mol/l NaHCO_3 and $1,8 \times 10^{-3}$ mol/l Na_2CO_3 .

5.5.2 Standard sulphate stock solution, $10,4 \times 10^{-3}$ mol/l SO_4^{2-}

Use a commercially available sulphate stock solution of 1 000 mg/l sulphate ($10,4 \times 10^{-3}$ mol/l) with a minimum content of 99,0 %. As an alternative prepare the standard solution as follows:

Dissolve 1,814 g of analytical grade potassium sulphate (K_2SO_4) in water (5.3) and dilute to a 1 000 ml volumetric flask. 1 ml of stock solution corresponds to 1 mg of SO_4^{2-} .

NOTE Calibration standards are prepared by diluting the standard stock solution with the absorption solution as specified in 9.2.3.2.

5.5.3 Regeneration solution for suppressor

For the exact composition of the suppressor regeneration solution, refer to the instructions given by the manufacturer of the suppressor.

NOTE An example is a solution of $12,5 \times 10^{-3}$ mol/l H_2SO_4 .

5.6 Reagent for Thorin analysis

5.6.1 2-propanol [$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$]

Use commercially available 2-propanol [$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$] in analytical grade (minimum content 99,8 %).

5.6.2 Barium perchlorate, standard volumetric solution, $[\text{Ba}(\text{ClO}_4)_2] = 0,005$ mol/l

Use a commercially available barium perchlorate solution (80 % 2-propanol/20 % water) with concentration of 0,005 mol/l $\text{Ba}(\text{ClO}_4)_2$, or prepare the standard solution as follows:

Dissolve 1,7 g of anhydrous barium perchlorate [$\text{Ba}(\text{ClO}_4)_2$] in about 200 ml of water in a 1 000 ml one-mark volumetric flask. Make up to the mark with 2-propanol (5.6.1) and mix well.

Titrate the solution accurately against a 0,005 mol/l standard volumetric sulphuric acid solution.

1 ml of exactly 0,005 mol/l barium perchlorate solution is equivalent to a mass of sulphur dioxide of 0,320 mg; ($f_a = 0,320$ mg/ml).

5.6.3 Potassium hydroxide, standard volumetric solution, [KOH] = 0,1 mol/l

Use a commercially available solution of 0,1 mol/l KOH in water.

NOTE This reagent is only necessary if the sampling gas contains high concentration of acid components (e.g. $\text{SO}_2 > 100$ mg/m³ or $\text{HCl} > 200$ mg/m³ or $\text{NO}_2 > 400$ mg/m³).

5.6.4 Perchloric acid, standard volumetric solution, [HClO_4] = 0,1 mol/l

Prepare an approximate 1 % solution of perchloric acid by mixing 16 ml of a commercially available solution of 60 % perchloric acid in water (5.3) and make up to 1 000 ml with water. Store this solution in a glass or PE bottle.

NOTE This reagent is only necessary if the sampling gas contains high concentrations of alkali components (e.g. $\text{NH}_3 > 50$ mg/m³).

5.6.5 Thorin, {4-[2-arsonophenyl]-azo]-3-hydroxy-2,7 naphthalene-disulfonic acid disodium salt} 2 g/l solution

Thorin is also known as Thoron or Thoronol, the sodium salt of 4-1(2-arsonophenyl)-azol-3-hydroxy-2,7 -naphthalene -disulfonic acid.

Dissolve 0,2 g of Thorin in water (5.3) in a 100 ml one-mark volumetric flask. Make up to the mark with water and mix well.

Store this solution in a bottle made of glass or polyethylene.

6 Sampling equipment

6.1 General

A known volume of flue gas is extracted representatively from a duct or a chimney during a certain period of time at a controlled flow rate. A filter removes the dust in the sampled volume, thereafter the gas stream is passed through a series of absorbers containing an absorption solution.

All parts of the sampling equipment upstream of the first absorber shall not react with or adsorb SO_2 (i.e. borosilicate glass, quartz glass, PTFE or titanium are suitable materials).

If an unheated gas connector line is used between the heated filter and the first absorber, it shall be thoroughly rinsed with fresh absorption solution after sampling and the rinsing solutions shall be combined with the sample.

An example of a suitable sampling train is shown in Figure A.1 in Annex A.

6.2 Sampling probe

In order to access the representative measurement point(s) of the sampling plane, probes of different lengths and inner diameters may be used, but the residence time of the sample gas in the probe shall be minimised.

The probe may be marked before sampling in order to demonstrate that the representative measurement point(s) in the measurement plane has (have) been reached.

The procedure described in 7.2.2 shall be used when the operator suspects that the flue gas is inhomogeneous.

When droplets are present in the flue gases, they may contain SO₂ dissolved in it. In that case, the probe is equipped with a nozzle and an isokinetic sampling shall be performed according to EN 13284-1.

NOTE 1 It is possible to perform the sampling of SO₂ and HCl simultaneously with the same probe without nozzle providing no droplets are present.

NOTE 2 SO₂ and H₂O can be measured simultaneously by using the same probe without nozzle and the same side stream absorbers, providing no droplets are present in the flue gases.

The sampling probe shall be surrounded by a heating jacket capable of producing a controlled temperature of at least 120 °C or 20 °C higher than the (acid) dew point of gases and shall be protected and positioned using an outer tube.

6.3 Filter housing

The filter housing may be located either:

- directly in front of the probe behind the entry nozzle (in-stack filtration); or
- directly behind the probe (out-stack filtration).

If condensation is liable to occur in the sampling probe or in the filter housing, then a heated out-stack filter housing shall be used. For out-stack filtration, the filter housing shall be heated at a controlled temperature of at least 120 °C or 20 °C higher than the (acid) dew point of gases. It shall be connected to the probe without any cold path between the two.

Filter housings of different designs may be used, but the residence time of the sample gas shall be minimised.

NOTE 1 The filter housing shall have the possibility to be jointed with the probe thereby avoiding leaks.

NOTE 2 A stop valve after the filter housing can be useful to prevent back flush of absorption solution into the probe or into the filter when sampling in flue gases under unfavourable conditions (e.g. high depression in the duct).

NOTE 3 In special cases where the sample gas temperature is > 200 °C, the heating jacket around the sampling probe, filter housing and connector line may be switched off. However the temperature in the sampled gas just after the filter housing, should not fall below 20 °C above the (acid) dew point temperature.

6.4 Particle filter

The filter shall have an efficiency better than 99,5 % on a test aerosol with a mean particle diameter of 0,3 µm, at the maximum flow rate anticipated, (or 99,9 % on a test aerosol of 0,6 µm mean diameter).

Glass filters cannot be used because glass reacts with SO₃, but ceramic, quartz or PTFE are suitable materials.

6.5 Temperature controller

A temperature controller is required for the filter housing. It shall be capable of controlling temperature with an uncertainty of ± 2,5 K or better.

6.6 Absorbers

6.6.1 General

For efficient absorption two absorbers shall be placed in series.

NOTE Downstream of these absorbers, an extra empty absorber may be used as a liquid trap and as a protection for the downstream equipment.

Some examples of absorbers are given in Annex B.

The absorption efficiency of the first absorber shall be better than 95 % or the concentration of sulphate ion in the second absorber shall be less than the detection limit. This efficiency shall be checked periodically at the normal flow rate according to the procedure described in 6.6.2. Except when the absorption efficiency shall be checked, the solutions from the two absorbers can be combined and analysed together.

NOTE Cooling of the absorbers may be useful to avoid excessive evaporation from the first absorption bottle.

6.6.2 Test of absorption efficiency

Place a suitable volume of absorption solution into each of the two absorbers, where the first absorber is denoted 1 and the second absorber 2. Assemble the apparatus to obtain a sampling train as illustrated in Figure A.1.

Carry out sampling in normal conditions. At the end of the sampling period, switch off the sampling pump, record the time and volume on the gas meter.

Remove the absorber from the sampling train and transfer the sample solutions from absorber 1 and 2 into two separate sample bottles. If a trap is used behind the absorbers to collect any solution carry-over, its contents shall be combined with the sample of bottle 2. Rinse each absorber with the absorption solution (5.4) thoroughly and particularly the fritted glass dividers to recover the absorption solution trapped in it and add the rinsing solutions to the appropriate absorber sample. Rinse all the unheated parts of the sampling system between the filter and the absorber 1 and add the rinsing to the content of bottle 1.

Analyse samples 1 and 2 as described in clause 9 to determine the sulphate content, q_{s1} and q_{s2} . Calculate the absorption efficiency ε of absorber 1 as follows:

Absorption efficiency of absorber 1 in % is $\varepsilon = q_{s1} \times 100 / (q_{s1} + q_{s2})$.

6.7 Sampling pump

Leak-free pump capable of sampling gas at a set flow-rate.

A sampling flow rate between 0,06 m³/h to about 0,2 m³/h allows to reach a good absorption with a low pressure of -10 kPa to -30 kPa.

NOTE 1 A small surge tank may be used between the pump and the rotameter to eliminate the pulsation effect of the diaphragm pump on the rotameter.

NOTE 2 A rotameter (optional) facilitates the adjustment of the nominal sampling flow-rate.

NOTE 3 A regulating valve (optional) is useful for adjusting the sample gas flow-rate.

6.8 Gas volume meter

Any dry or wet gas volume meter may be used providing the volume is measured with a relative uncertainty not exceeding ± 2 % at actual conditions.

The gas volume-meter shall be equipped with a temperature measuring device (uncertainty of calibration less than $\pm 2,5$ K). The absolute pressure at the gas volume meter (uncertainty of calibration less than $\pm 1,0\%$) can be determined from the relative pressure and the ambient pressure.

When using a dry gas volume meter, a condenser and/or a gas drying system shall be used which can lead to a residual water vapour content of less than 10 g/m^3 (equivalent to a dew point of $10,5^\circ\text{C}$ or a volume content $\chi(\text{H}_2\text{O}) = 1,25 \%$).

NOTE For example a glass cartridge or absorption bottle packed with silica gel (1 mm to 3 mm particle size), which has been previously dried at least at 110°C for at least 2 h.

When using a wet gas volume meter, a correction shall be applied for water vapour, to obtain a dry gas sampled volume.

7 Sampling procedure

7.1 General requirements

The sample and the point of measurement shall be representative of the emission of the process.

The following points should be considered when sampling:

- nature of the plant process e.g. steady state or discontinuous. If possible the sampling and measuring programme should be carried out under steady operating conditions at the plant;
- expected concentration to be measured and any required averaging period, both of which can influence the measuring and sampling time.

The minimum sampling period is 30 min. When the expected mass concentration of SO_2 is low, the sampling period or the sampling volume may be extended.

7.2 Preparation and installation of equipment

7.2.1 Sampling location

The sampling location chosen for the measurement devices and samplings shall be of sufficient size, easily passable and made in such a way that an emission measurement representative for the measurement task and technically perfect is possible. In addition, the sampling location shall be chosen with regard to safety of the personnel, accessibility and availability of electrical power.

7.2.2 Sampling point(s)

It is necessary to ensure that the gas concentrations measured are representative of the average conditions inside the waste gas duct. In most cases, a single sampling point situated in the middle of the duct shall be selected. For larger ducts, this point can be situated closer to the sampling port but not too close to avoid any disturbance of the flow or concentration due to influences from the sampling port.

However, when non-homogeneity of the flue gas is suspected, it shall be demonstrated if the flue gas is homogeneous or not. The homogeneity can be demonstrated with a continuous measurement of O_2 or CO_2 using a fixed and a moving probe. Sampling in non-homogeneous flue gases requires following relevant standards or technical specifications proposed by CEN TC 264.

When droplets are present in the flue gases, then the probe is equipped with a nozzle and an isokinetic sampling shall be performed at the different points required by EN 13284-1.

7.2.3 Assembling the equipment

Prepare all the glassware, filters and absorption solutions required for the sampling equipment. Assemble the whole sampling train (an example is given in Annex A). Special care shall be taken during the rinsing of absorbers with fritted gas dividers, as it is difficult to completely recover the absorption liquid used for rinsing these devices. Consequently rinsing needs to be thorough.

The dead volume in the sampling probe, filter housing and tubing up to the first absorber should be minimised.

The two absorbers shall be filled with the appropriate amount of the absorption solution (5.4) to guarantee a sufficient gas/liquid contact. An empty absorber may be added behind the second one as a protection for the downstream suction equipment.

7.2.4 Heating of the sampling line

Maintain the temperature of the sampling probe and particle filter housing up to included at a temperature of at least 120 °C or 20 °C above the (acid) dew-point temperature of the sample gas.

7.2.5 Leak test

Perform a leak test on the sampling train. Check the sampling line for leakage according to the following procedure or any other relevant procedure:

- assemble the complete sampler system, including charging the filter housing and absorbers;
- seal the nozzle inlet;
- switch on the pump(s);
- after reaching minimum pressure read the flow rate;
- leak flow rate shall be measured (e.g. by a rotameter) and shall not exceed 2 % of the expected sample gas flow rate.

Perform the leak test at the operating temperature unless this conflicts with safety requirements.

Integrity of the sampling system can be also tested during sampling by continuously measuring the concentration of a suitable stack gas component (e.g. O_2) directly in the stack and downstream the sampling line. Any systematic difference between those concentrations indicates a leak in the system.

7.3 Performing sampling

7.3.1 Introduction of the probe in the duct

After the leak test, insert the sampling probe through the sampling port and place the probe at the representative point(s) in the measurement plane (see 7.2.2). Seal the resultant space around the sampling probe and the access hole with an appropriate material such that ambient air is not induced into the duct (nor does any flue gas escape from the duct). In cases where large pressure differences exist between the ambient air and sample gas, care shall be taken that the absorption solution does not leave the absorber.

Control the jacket and filter housing temperatures.

7.3.2 Sampling

Record the reading on the gas meter V_1 and the time.

Start the sampling pump and adjust the regulating valve to give the desired sample gas-volume flow rate. The selected sample gas flow-rate shall be kept within $\pm 10\%$ of the nominal flow rate (see NOTE 2). However, when non-homogeneity is suspected, it shall be demonstrated if the flue gas is homogeneous or not. Sampling in non-homogeneous flue gases shall follow the requirements of relevant standards or technical specifications proposed by CEN TC 264.

Record the reading on the gas meter thermometer T_j and the absolute pressure P_m on the gas meter at least 5 min after starting sampling and at the end of the sampling period.

NOTE 1 For a sampling train as shown in Figure A.1 in Annex A, with gas meter placed downstream pump, absolute pressure on the gas meter is close to atmospheric pressure.

The minimum sampling period is 30 min. However, when the expected mass concentration of sulphur dioxide is low, the sampling period may be extended to have analytical results above the detection limit of analysis.

NOTE 2 The sampling flow rate can be derived from the information given on the sampled volume of gas given in Table 1 (see 9.3.2), and the chosen sampling period (i.e. 30 min).

NOTE 3 It is recommended to perform a leak test after sampling when a grid measurement has been performed.

7.3.3 Rinsing of the sampling system and preparation of the samples

At the end of the sampling period, switch off the sampling pump (care shall be taken that the absorption solution does not leave the absorber), record the time and the reading on the gas meter V_2 . Remove the absorber from the sampling train and quantitatively transfer both sample solutions into a bottle of suitable capacity or into two separate bottles if the absorption efficiency shall be checked. The bottles shall be made of an appropriate material.

Rinse thoroughly all the parts of the absorbers, and particularly the fritted glass dividers with absorption solution. Force the solution through the fritted glass dividers, if they exist, using a pressure bulb. If used, the trap shall also be rinsed and the rinsing solution combined with the other rinsing solutions. Add the rinsing solution to the corresponding sample solution in the sample bottle.

The volume of the absorption solution including the rinsing solution is determined with an uncertainty lower than $\pm 1\%$ rel.

All the unheated parts from the filter housing until the second absorber in the sampling train shall be rinsed.

7.4 Measuring series

To take further flue gas samples, place the appropriate volume of absorption solution into each of the two absorbers, assemble the sampling train and proceed as specified above.

7.5 Field blank

To check the sampling procedure, a field blank shall be performed at least before each measurement series or at least once a day, following the whole measurement procedure specified in this European Standard and including the sampling procedure described in 7.2.3 to 7.3 without the suction step i.e., without starting and operating the suction device.

The average sampling time shall be used for calculation of the blank value expressed in mg/m^3 .

If the equipment in contact with the measured substance is cleaned and reused in the field, a field blank shall also be taken after the measurement series. If several measurements are performed at the same industrial process or on several lines of the same industrial process, then only one single field blank at the beginning and one at the end of the series shall be performed.

The field blank shall be less than 10 % of the emission limit value (ELV). If the calculated value of the measurement is less than the field blank, the measured value result shall be reported as less or equal to the field blank.

The result of field blank shall not be subtracted to the result of measured value. However, it is necessary to take into account the value of the field blank in the calculation of the uncertainty of the measured value (see Annex C).

7.6 Absorption efficiency

If no prior experience is available which verifies the absorption efficiency for the chosen sampling equipment arrangements and sampling conditions, then the procedure specified in 6.6.2 shall be followed at the measurement site.

8 Analysis equipment

8.1 Ion chromatograph

Analysis equipment consists of a complete analytical system with an ion chromatograph and all required accessories including syringes, analytical columns, compressed gases, a detector and a recording device. The essential minimum requirements for an ion chromatograph system for the scope of this European Standard are as follows:

- a) sample injection system: constant-volume injection system shall be used;
- b) anion separator pre-column (optional): this column is used to protect the measuring device;
- c) anion separator column: this column produced the separation of anions in the sample, enabling a clear measurement of the sulphate anion peak area and height. Other anions, which may be present in the sample solution, include F, Cl⁻ and NO₃⁻. The resolution power of the separator column shall be sufficient to ensure that the peak resolution R_s shall not fall below 1,3 where:

$$R_s = \frac{2(t_2 - t_1)}{w_1 + w_2} \quad (1)$$

where

- t_1 retention time of the first peak, in seconds;
- t_2 retention time of the second peak, in seconds;
- w_1 peak width on the time axis, of the first peak, in seconds;
- w_2 peak width on the time axis, of the second peak, in seconds.

In general it is recommended to use a pre-column or an anion guard column to protect the anion separator column. If this is omitted from the system, the retention times shall be shorter. Two different types can be used: those containing the same substrate as the separator column, and those packed with a macro porous polymer.

- d) detector: the method of detection should rely on a measurement of the electrical conductivity with or without a suppressor device;
- e) recording device: a system using a strip charts recorder and integrator or another computer-based data system is acceptable.

8.2 Thorin method

8.2.1 General

To determine the titration end point, a photodiode in connection with a titrator or a spectrophotometer can be used.

8.2.2 Spectrophotometer (optional)

Use a spectrophotometer with single or double light beam.

When a spectrophotometer with one single light beam is used, it is necessary to be able to tune the incident flux for example, by adjusting a diaphragm or a slot width. The best accuracy is reached when the incident flux can be adjusted to give an absorbance value as close as possible to zero for the highest concentration to be measured. However, the best result is usually obtained when the adjustment of the incident flux gives an absorbance reading of 0,800 for the chemical blank when using an optical cuvette of 20 mm and a value of 0,400 for a cuvette of 10 mm.

8.2.3 Glass optical cuvette

Glass optical cuvette of 10 mm or 20 mm optical length. If more than one cuvette is used, then they shall be photometrically matched.

9 Analytical procedure

9.1 General

After sampling, the solutions shall be analysed by one of the following two methods:

— method A: Ion Chromatography;

Analytical detection limit L_D : equal or above 0,05 mg SO_4^{2-} per litre of absorption solution (corresponding to concentrations equal or above 0,1 mg/m³ SO_2 in flue gas with half an hour sampling in 100 ml absorption solution and 1 l/min flow-rate).

— method B: Thorin Method.

Analytical detection limit L_D : equal or above 1 mg SO_4^{2-} per litre of absorption solution (corresponding to concentrations equal or above 2,2 mg/m³ SO_2 in flue gas with half an hour sampling in 100 ml solution and 1 l/min flow-rate).

The solutions that shall be analysed are:

- a) field blanks;
- b) samples.

9.2 Ion Chromatography method

9.2.1 General procedure

Start up the ion chromatograph in accordance with the manufacturer's instructions and confirm the readiness for operation (e.g. stable baseline). Check the calibration daily according to 9.2.3.4 and, if necessary, recalibrate as described in 9.2.3.2 and 9.2.3.3.

Before each analysis, the exact total volume of the sample v_s (absorption solution and rinsing solution) is determined using a graduated measuring cylinder (accurate to 1 ml). This may be done gravimetrically taking the density of the solution into account or by transferring the sample into a measuring flask of suitable size and making up to the mark with unused sorption solution by titration.

Load and inject a fixed amount of sample solution. After each analysis, flush the injection loop thoroughly between each sample run using the eluent solution. Use the same size loop for calibration standards and samples. Record the resulting output signal, i.e. sulphate peak size, in units of area or height.

NOTE A filtration of the sample can be necessary.

The width of the retention-time window used to make identifications should be based upon measurements of actual retention-time variations of calibration standards employed over the course of a day. Three times the standard deviation of a retention time or 10 % of the retention time can be used to calculate a suggested window size. However, the experience of the analyst should weight heavily in the interpretations of the chromatograms.

If the output signal exceeds the calibrated working range of the system, dilute the sample with an appropriate amount of absorption solution and reanalyse. If the resulting chromatogram fails to produce adequate resolution, or the identification of the sulphate peak is questionable, fortify the sample with an appropriate amount of standard and reanalyse.

Compute sample sulphate concentration q_s in milligrams per litre of solution by comparing sample output signal with the sulphate calibration curve, taking into account any dilution factor applied.

Similarly, determine the sulphate concentration of the field blank and of the chemical blank. The chemical blank value shall usually be evaluated each time the absorption solution is prepared.

9.2.2 Interferences

Interference may be caused by contaminants in the reagent, in water and in all the points of the chromatography system such that this leads to high baseline noise or interfering signals.

9.2.3 Calibration

9.2.3.1 Set up of the chromatograph

Establish ion chromatographic operating parameters equivalent to those given in the manufacturer's instructions.

9.2.3.2 Preparation of calibration standards

Prepare calibration standards at a minimum of three concentration levels over a chosen working range and a zero sample (corresponding to a sample of absorption solution). A calibration standard consists of a sulphate solution prepared in the laboratory from the standard sulphate stock solution (5.5.2) and accurately diluted as needed with absorption solution. The standards are used to calibrate the output signal with respect to the sulphate concentration and to establish a corresponding calibration curve.

If the calibrated working range exceeds the linear range of the system, a sufficient number of calibration standards shall be analysed to allow an accurate calibration curve to be established. One of the calibration standards should be at a concentration near, but above, the detection limit. The other calibration standards should correspond to the range of concentrations expected in the samples or they should define the working range of the detector if the system is operated on a number of attenuation ranges. Each range shall be calibrated individually unless the attenuation range settings are proven to be linear, (i.e. 2 %).

NOTE As an example, for sulphur dioxide concentrations in the range 7 mg/m³ to 333 mg/m³, with half an hour sampling in 100 ml solution and 1 l/min flow-rate, a working range for sulphate concentrations between 3 mg/l and 150 mg/l is suitable.

9.2.3.3 Calibration of the chromatograph

Inject an appropriate amount of the first calibration standard depending on the injection loop volume. Tabulate the output signal against concentration. Flush the injection loop with eluent solution and repeat for the other calibration standards and zero sample. The results are used to prepare a sulphate calibration curve. Record the retention times during this procedure.

9.2.3.4 Calibrations checks

Verify the calibration curve each working day, whenever the anion eluent is changed or after every 20 samples. Reference material shall be used to check the calibration regularly. In addition, if the retention time varies from the expected values by more than $\pm 10\%$, repeat the test using fresh calibration standards. Then, if the results are still more than $\pm 10\%$, prepare a new sulphate calibration curve.

9.2.3.5 Measuring range

Non-linear response within the range of the detector can be noticed when the separator column capacity is exceeded (overloaded). The output signals should be compared for an undiluted sample and for a sample when diluted 1 to 10. If the calculated sulphate concentrations of the two samples are the same, samples with this sulphate concentration do not need to be diluted.

9.3 Thorin Method

9.3.1 Pre-treatment of sample solution before analysis for Thorin method

Measure and monitor the pH of the absorption solution used for sampling using the direct reading pH meter. Adjust the pH of the sample solution to pH 3,5 by adding the appropriate volume of the standard volumetric potassium hydroxide solution (5.6.3) or by adding the standard volumetric perchloric acid solution (5.6.4) as required. Transfer the resultant solution into a volumetric flask of a suitable nominal capacity. Make up to the mark with water and mix well.

The direct reading pH meter should preferably be equipped with a temperature compensation and shall have a measuring range 0 to 14. The limits of error shall be $< 0,2$ units at pH 3,5.

Calibrate the direct reading pH meter using an appropriate buffer solution. The pH of the buffer solution shall be accurately known at a given temperature. After the calibration, rinse the electrodes thoroughly before use.

9.3.2 General procedure

Transfer the aliquot specified in table 1 of the "pre-treated sample solution" obtained as described below, into a cortical flask of suitable nominal capacity and add 80 ml of 2-propanol (5.6.1) plus four drops of the Thorin solution (5.6.5) and mix well. Record the ratio, f_v , of the volume of the total sample solution to the volume of the aliquot taken.

If the absorption efficiency is checked, treat each absorber solution with its rinsing solution separately.

Table 1 — Example of sampling parameters and analytical solution requirements for typical SO₂ concentrations

Expected mass concentration of sulphur dioxide mg/m ³	Capacity of the absorbers ml	Volume of the absorption solution to be placed into each absorbers ml	Volume of the sample gas to pass through the sampling train m ³	Volume of the pre-treated sample solution ml	Aliquot of the pre-treated sample solution to be titrated ml	Volume of Ba(ClO ₄) ₂ solution used for the titration the aliquot ml
30 to 100	100	40	0,100	100	20	1,875 to 6,5
100 to 500	100	40	0,060	100	20	3,75 to 18,75
500 to 1 000	250	80	0,060	250	20	7,5 to 15
1 000 to 2 000	250	80	0,030	250	20	7,5 to 15
2 000 to 5 000	250	80	0,030	250	20	15 to 37,5

Use a burette with a spindle-valve or an automatic titrator. Titrate the solution obtained with the standard volumetric barium perchlorate solution (5.6.2) until the end-point is reached (the colour changes from orange-yellow to permanent light pink). Record the titration volume. Repeat this procedure and, if the end-point is indistinct, repeat a third time and calculate the mean of significant values.

NOTE In some types of light, the change in colour from orange-yellow to permanent light pink is very difficult to observe, as for example in fluorescent lighting and direct sunlight. Therefore it is preferable to titrate under a daylight lamp or using a photometer with a glass fibre-optic light guide capable of measuring the transmission of light at a wavelength of 520 nm or 481 nm, and an automatic titrator.

9.3.3 Preparation of a chemical blank solution

Prepare a chemical blank solution by adding 80 ml of 2-propanol (5.6.1) and four drops of the Thorin solution (5.6.5) to an aliquot of the absorption solution. The volume of the absorption solution used should correspond to the volume of the aliquot of the pre-treated sample solution chosen to be titrated. Analyse the chemical blank solution as specified above. The chemical blank value shall be evaluated each time the absorption solution is prepared.

9.3.4 Interferents

9.3.4.1 Anions

Anions that are absorbed by the absorption solution and that also form sparingly soluble salts with barium ions at pH 3,5 can interfere.

NOTE Interferences by inorganic gaseous chlorine and fluorine compounds and stable oxides are not expected in normal flue gases. In exceptional cases (extremely high flue-gas temperature) interferences by volatile sulphates and volatile salts of polyvalent metal cations can take place.

9.3.4.2 Volatile salts of polyvalent metal cations

Those which respond to the indicator Thorin. The metal cations can be removed from the treated joint sample solution by percolating the latter through a cation exchanger.

10 Determination of the characteristics of the method: sampling and analyse

10.1 General

When this European Standard is used as the SRM, the user shall demonstrate that:

- performance characteristics of the method given in Table 2 are better than the minimum performance criteria; and
- overall uncertainty calculated by combining values of selected performance characteristics by means of an uncertainty budget is less than $\pm 20,0$ % at the daily emission limit value.

The values of the selected performance characteristics shall be evaluated:

- for sampling step: by means of laboratory tests in order to determine uncertainty of the calibration of the equipment and by means of field tests in order to determine other parameters;
- for analytical step: by means of laboratory tests.

10.2 Relevant performance characteristics of the method and performance criteria

10.2.1 General

The uncertainty of the measured values is influenced by:

- sampling line and conditioning system;
- sampling procedure;
- analysis;
- site specific conditions.

The uncertainty budget is drawn up according to 10.3.

10.2.2 Sampling procedure

Table 3 gives an overview of the relevant performance characteristics and performance criteria, which shall be determined during laboratory and field tests according to the relevant CEN procedures for sampling.

Table 2 — Minimum performance characteristics of the sampling system

Performance characteristic	Lab. Test	Field test	Performance criterion
Sampling			
Determination of the volume of the absorption solution		X	$\leq \pm 1,0$ % of the volume of solution
Volume gas meter:			
— uncertainty of sample volume ^b	X ^a		$\leq \pm 2,0$ % of the volume of gas sampling ^a
— uncertainty of temperature ^b	X ^a		$\leq \pm 2,5$ K ^a
— uncertainty of absolute pressure ^b	X ^a		$\leq \pm 1,0$ % of the absolute pressure ^a
Absorption efficiency ^c		X	> 95 %
Leak in the sampling line		X	$\leq 2,0$ % of the nominal flow rate
Value of the field blank		X	$\leq 10,0$ % of ELV
^a Performance criteria corresponding to the uncertainty of calibration. ^b The uncertainty of the sampled volume is a combination of uncertainties due to: calibration, drift (random drift, drift between 2 calibrations), resolution, and reading. The uncertainty of temperature and absolute pressure at the gas volume meter is a combination of uncertainties due to: calibration, drift (random drift, drift between 2 calibrations), resolution, reading, and repeatability. ^c This characteristic is an assurance quality check to quantify the absorption efficiency in the first absorber; but it does not quantify a possible loss of absorption, and therefore it is not included in calculation of overall uncertainty.			

10.2.3 Analyse procedure

10.2.3.1 Sources of uncertainty

Main possibly sources of uncertainty associated to analysis are:

For analysis by ion chromatography:

- performance characteristics of the analysis equipment;
- preparation of calibration standards: purity of stock standard solution, and ratio of dilutions;
- linearity of calibration curve depending on the extend of working range;
- measurement of volume of aliquot solution injected for analyse (ratio of the total absorption solution volume and the volume of the aliquot taken for injection);
- if a dilution of the absorption solution is necessary before analyse: ratio of dilution;
- interferences;
- drift of retention time;
- repeatability.

For analyse by titration:

- performance characteristics of analytical equipment;
- preparation of standard volumetric solution (5.6.2) used for titration;

- adjustment of pH of absorption solution;
- measurement of volumes of aliquots of titrated solutions (ratio of the volume of the sample solution to be titrated and the volume of the aliquot taken): for the absorption solution and chemical blank;
- detection of the colour change;
- measurement of volume of standard volumetric solution used for titration;
- interferences;
- repeatability.

10.2.3.2 Performance criterion of analysis

Because all the components of uncertainty attached to the analysis are difficult to identify and to estimate, the laboratory can determine the overall uncertainty due to analysis by taking the standard deviation of repeatability calculated during an interlaboratory comparison. A maximum performance criteria is given in the following Table 3.

One example of analytical validation is presented in Annex D. The exercise has been made through a round robin test consisting of sulphate analysed in 4 solutions by ionic chromatography and by Thorin method. Twenty-two laboratories took part to the test. The round robin test allowed to determine the standard deviation of reproducibility.

Table 3 — Performance characteristics of analytical procedure

Performance characteristic	Lab. Test	Field test	Performance criterion
Standard deviation of repeatability of sulphate ions analyse	X		$\leq 2,5 \%$ of the measured value (value of quantity of sulphate ions in the solution; in $\text{mg SO}_4^{2-} / \text{l}$ of solution)

10.3 Establishment of the uncertainty budget

An uncertainty budget shall be established to evaluate whether or not the method fulfils the requirements for a maximum allowable overall uncertainty.

The overall uncertainty for this method used as a reference shall be lower than $\pm 20,0 \%$ at the daily emission limit value. This overall uncertainty is calculated on dry basis and before correction to the O_2 reference concentration.

The principle of calculation of the overall uncertainty is based on the law on propagation of uncertainty laid down in the ENV 13005:

- determine the standard uncertainties attached to the performance characteristics to be included in the calculation of the budget uncertainty by means of laboratory and field tests, and according to ENV 13005;
- calculate the uncertainty budget by combining all the standard uncertainties according to ENV 13005;
- values of standard uncertainty that are less than 5 % of the maximum standard uncertainty can be neglected;
- calculate the overall uncertainty at the daily emission limit value, on dry basis.

An example of the evaluation of overall uncertainty is given in Annex C.

11 Expression of results

Calculate the mean value T_m of the temperature at the gas volume meter:

$$T_m = \frac{(T_1 + T_2 + \dots + T_j + \dots + T_n)}{n} \text{ (in K)} \quad (2)$$

where

n is the number of temperature readings;

T_j taken during the sampling.

Calculate the dry gas volume sampled from the gas meter, at standard conditions, in cubic metre $V_{m(std)}$:

$$\text{If a dry-gas volume meter is used } V_{m(std)} = (V_2 - V_1) \times \frac{T_{std}}{T_m} \times \frac{P_m}{P_{std}} \quad (3)$$

$$\text{If a wet-gas volume meter is used } V_{m(std)} = (V_2 - V_1) \times \frac{T_{std}}{T_m} \times \frac{P_m - P_{sat}(T_m)}{P_{std}} \quad (4)$$

where

T_m is the mean temperature (in Kelvin) of the sampled gas at the gas-meter

T_{std} is the standard temperature, 273 K

P_m is the absolute pressure (in kilopascals) at the gas volume meter

$P_{sat}(T_m)$ is the saturation vapour pressure (in kilopascals) of water at temperature of the volume gas meter

P_{std} is the standard pressure, 101,3 kPa

$V_2 - V_1$ is the sampled gas volume (in cubic metre)

Calculate the mass concentration of sulphur dioxide C_m in mg/m^3 present in the sampled gas at the standard gas conditions, related to the dry gas, using equation (3) or (4) depending on the analytical method applied:

For the ion chromatography method:

$$C_m = \frac{(q_s - q_{cb}) \times V_s \times \frac{64,1}{96,1}}{V_{m(std)}} \text{ (mg/m}^3\text{)} \quad (5)$$

where

q_s is the mass concentration of sulphate (in milligrams per litre) collected in the sample absorption solution

v_s is the volume of the absorption solution (in litres)

$V_{m(std)}$ is the volume of dry gas sampled from the gas meter, at standard conditions (in cubic metres)

NOTE 1 When the solution from each absorber (1 and 2) and the associated rinsings are analysed individually and the concentrations and volumes of these solutions are denoted as q_{s1} , q_{s2} , v_{s1} , v_{s2} respectively, then the term $(q_s \cdot v_s)$ is replaced by $[(q_{s1} \cdot v_{s1}) + (q_{s2} \cdot v_{s2})]$.

For the titration method:

$$C_m = \frac{f_a \times f_v \times (S_s - S_{cb})}{V_{m(std)}} \quad (6)$$

where

- f_a is the equivalent mass of sulphur dioxide (in milligram per millilitre) of 1 ml of the titration solution (barium perchlorate standard volumetric solution (5.6.2))
- f_v is the ratio of the volume of the pre-treated sample solution to the volume of the aliquot taken for the titration
- S_s is the volume (in millilitres) of titration solution used for the titration of the aliquot of the pre-treated sample solution
- S_{cb} is the volume (in millilitres) of titration solution used for the titration of the aliquot of the chemical blank solution

NOTE 2 When the solution from each absorber (1 and 2) and the associated rinsings are analysed individually and the volumes of titration solutions used are denoted as S_{s1} , S_{s2} respectively, then the term $(S_s - S_{cb})$ is replaced by $(S_{s1} + S_{s2} - 2S_{cb})$ if f_v is the same for both partial solutions.

Concentrations are generally expressed at a reference concentration of O_2 defined in the European Directives:

- 11 % for incineration of waste in waste incinerators;
- 10 % for co-incineration of waste in cement kilns;
- 3 % for combustion of gas or liquid fuels;
- 6 % for combustion of solid fuels;
- 15 % for gas turbines.

The corrected concentration of measurand C_{corr} is calculated using the followed equation:

$$C_{corr} = \frac{21 - O_{2,ref}}{21 - O_{2,mes}} \times C_{actual} \quad (7)$$

where

- C_{corr} is the corrected concentration of measurand
- C_{actual} is the measured concentration of measurand at actual O_2 concentration
- $O_{2,mes}$ is the measured mean dry oxygen content during the sampling time
- $O_{2,ref}$ is the oxygen reference concentration

12 Evaluation of the method in the field

The method has been evaluated during six field tests, on waste incineration installations, co-incineration installations and large combustion plants. Each test was performed by at least four different European measuring teams originating from ten member countries.

The characteristics of installations, the conditions during field tests and the values of repeatability and reproducibility in the field are given in Annex E.

13 Equivalence of Thorin and ion chromatography methods

13.1 General

This equivalence has been checked according to the Technical Specification CEN TS 14793.

An example of the implementation of the method to check the equivalence of both methods described in this European Standard is given in Annex E.

13.2 Range

The range of equivalency begins at the limit of quantification L_Q of the Thorin method, which is between 3 mg per litre SO_4^{2-} and approximately 10 mg per litre SO_4^{2-} (according to the experience of the laboratory).

13.3 Matrix effect

The equivalency has been checked for waste incinerators, co-incinerators and large combustion plants.

13.4 Comparison of repeatability and trueness

The standard deviation of repeatability s_r , expressed in mg/m^3 for this reference method is:

- ion chromatography method $S_r(C) = 0,033\ 8\ C + 1,5$ where C is the concentration in mg/m^3 ;
- Thorin method $S_r(C) = 0,027\ 3\ C + 2,4$ where C is the concentration in mg/m^3 ;
- standard deviation of reproducibility expressed in mg/m^3 (n) for this reference method is s_r ;
- ion chromatography method $S_r(C) = 0,051\ 4\ C + 2$ where C is the concentration in mg/m^3 ;
- Thorin method $S_r(C) = 0,043\ C + 3,4$ where C is the concentration in mg/m^3 .

14 Equivalence with an alternative method

In order to show that an alternative method is equivalent to the reference method, follow the procedures described in the CEN TS 14793.

The maximum allowable standard deviation of repeatability and reproducibility expressed in mg/m^3 for this reference method are:

$$S_{r \text{ limit}}(C) = 0,051 C + 2,3 \quad (8)$$

$$S_r(C) = 0,051 4 C + 2 \quad (9)$$

where

C is the concentration in milligrams per cubic metre.

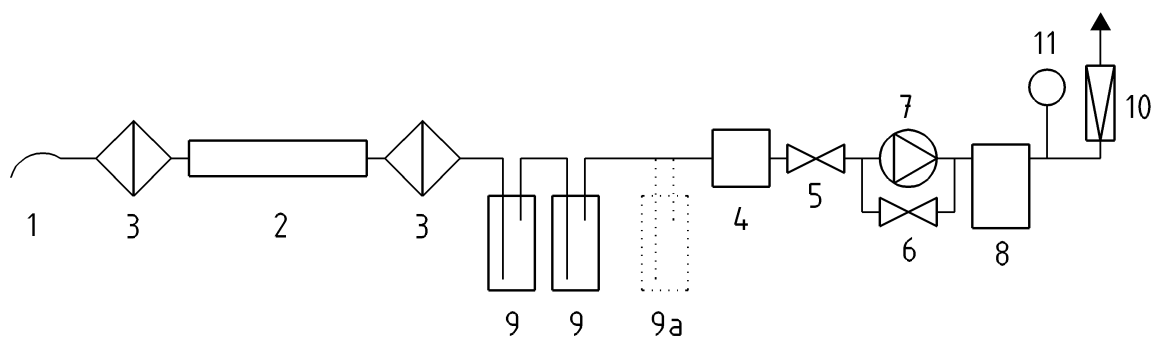
15 Test report

The test report shall include the following information:

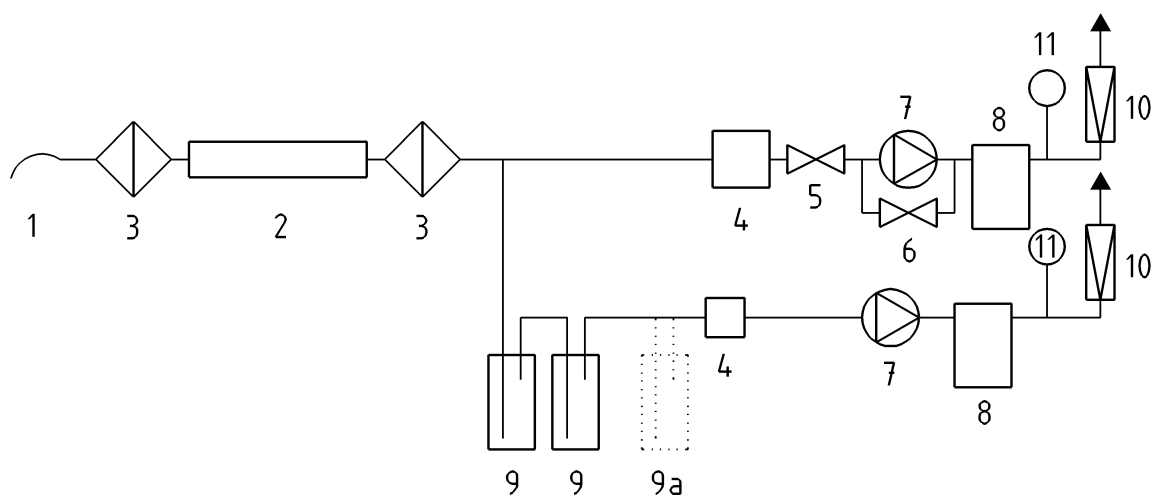
- a) reference to this European Standard;
- b) description of the purpose of tests;
- c) information about the personnel involved in the measurement;
- d) description of plant and process;
- e) description of the operating conditions of the plant process;
- f) identification of the sampling plane;
- g) description of the location of the sampling point(s) in the sampling plane;
- h) changes in the plant operations during sampling, for example burner changes;
- i) characteristics of sampling equipment;
- j) number of tests;
- k) for each test: date, time, duration and identification of samples(s);
- l) test results: sampling volume, concentrations;
- m) blank values;
- n) absorption efficiency;
- o) actual uncertainty budget established according to EN ISO 14956 or ENV 13005;
- p) any deviations from this European Standard.

Annex A (informative)

Type of sampling equipment



a) Main-stream sampling



b) Side-stream sampling

Key

- | | |
|---|---|
| 1 Nozzle | 7 Pump |
| 2 Probe | 8 Gas volume meter |
| 3 Filter (either behind or in front of the probe) | 9 Absorber |
| 4 Dryer cartridge | 9a Safety bottle (trap) |
| 5 Valve | 10 Gas flow meter |
| 6 By-pass valve | 11 Temperature and pressure measurement |

Figure A.1 — Types of sampling equipment

Annex B (informative)

Examples of absorbers

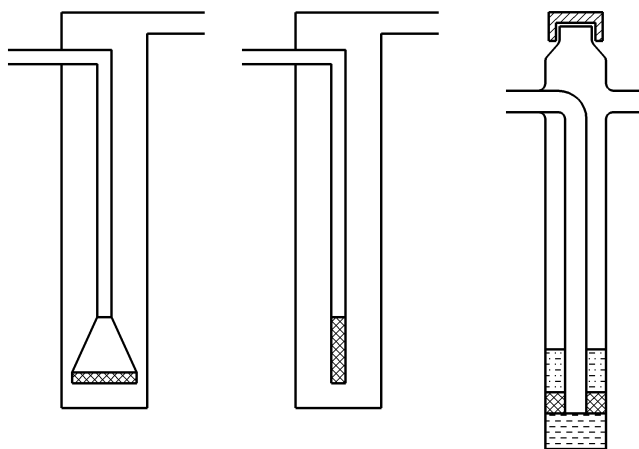


Figure B.1 — Examples of absorbers with fritted gas divider (0,03 m³/h to 0,18 m³/h)

NOTE For high absorption efficiency it is advisable to distribute the gas stream in the absorption liquid as homogeneously as possible and have reasonably long contact time between gas and liquid. In practice a sufficient free space should break down any foam, which may be formed while the gas is bubbling through the liquid.

Annex C (informative)

Example of assessment of compliance or reference method for SO₂ with requirements on emission measurements

C.1 General

This informative annex gives an example of calculation of the overall uncertainty established with configuration 1.

C.2 Process of uncertainty estimation

C.2.1 General

The procedure for calculating measurement uncertainty as follows is based on the law on propagation of uncertainty laid down in EN ISO 14956 or ENV 13005. The calculation procedure presents different steps:

C.2.2 Determination of the model equation

Define the measurand and all the parameters that influence the result of the measurement. These parameters, called “input quantities” shall be clearly defined.

Identify all sources of uncertainty contributing to any of the input quantities or to the measurand directly.

Then the model equation, that is to say the relationship between the measurand and the influence quantities, shall be established, if possible in mathematical equation form.

C.2.3 Quantification of uncertainty components

Each uncertainty source is estimated to obtain its contribution to the overall uncertainty.

Use available performance characteristics of the measurement system, data from the dispersion of repeated measurements, data provided in calibration certificates....

Convert all uncertainty components (e.g. performance characteristics) to standard uncertainties of input and influence quantities.

C.2.4 Calculation of the combined uncertainty

Then the combined uncertainty u_c is calculated by combining standard uncertainties, by applying the “law of propagation of uncertainty”.

In general, the uncertainty associated to a concentration is expressed in overall uncertainty form. The overall uncertainty U_c corresponds to the expanded combined uncertainty, obtained by multiplying by a coverage factor “ k ”: $U_c = k u_c$. The value of the coverage factor k is chosen on the basis of the level of confidence required. In most cases, k is taken equal to 2, for a level of confidence of approximately 95 %.

The following Tables give one example of:

- specific conditions of the site (Table C.1), that is to say the values and the variation range of the influence parameters;
- performance characteristics of the method (Table C.2) related to the parameters which can have an influence on the response of the analyser: the performance characteristics of the analyser and the effect of influence parameters (e.g. chemical interferences, and environmental conditions such as ambient temperature, voltage and pressure).

C.3 Specific conditions in the field

The mass flow is homogenous.

Table C.1 — Example of measurement conditions

Specific conditions	Value / range
Studied concentration of SO ₂ (limit value of SO ₂ for the site, at $O_{2,ref}$)	50 mg/m ₀ ^{3 a} at standard conditions of temperature and pressure, and at 11 % O ₂ concentration corresponding to: $q_s = 14,56 \text{ mg SO}_4^{2-} / \text{l}$ of absorption solution
O ₂ reference concentration for the site: $O_{2,ref}$	11 % volume
O ₂ measured concentration: $O_{2,mes}$	12,3 % volume $\pm$ 6% relative (k = 2)
Volume of solution V_s used for measurement	200 ml (total volume in 2 absorbers)
Volume of gas sampled	0,049 m ³
Mean temperature (in Kelvin) at the gas meter ^b	296,2 K
Mean absolute pressure at the gas meter ^c	100 281 Pa
Analyse	Ion chromatography
^a m ₀ ³ : cubic metre at standard conditions of temperature (273 K) and pressure (101,325 kPa). ^b Mean temperature is calculated from data recording of continuous temperature measurement (1 measurement / 30 s = 60 measurements in 30 min). The standard deviation of the mean of measurements calculated is equal to 0,231 K. ^c Mean absolute pressure is calculated from 5 measurements of relative pressure at the gas meter and 1 measurement of atmospheric pressure during the sampling period: — barometric pressure: 100,212 kPa.	

Table C.2 — Example of measured values of relative pressure

Measurement	1	2	3	4	5	Mean	Standard deviation of the mean of measured values (Pa)
Relative pressure at the gas meter (Pa)	70,5	68,7	69,0	68,6	69,8	69,2	0,287
Mean absolute pressure: 100,281 kPa.							

C.4 Performance characteristics of the method

Table C.3 — Example of performance characteristics

Performance characteristics for the reference method	Performance criteria	Laboratory or field tests results
Volume of absorption solution V_s: The total volume of solution is determined with a graduated tube: — tolerance of the tube; — reading; — standard deviation of repeatability of the measurement of the volume.	$\leq \pm 1,0$ % of the volume of solution	$\pm 1,4$ ml 2 ml 0,5 ml
Gas volume sampled $V_{m, std}$: — overall uncertainty of calibration; — standard deviation of repeatability of measurement; — drift between 2 adjustments; — reading.	$\leq \pm 2,0$ % of the measured value	$\pm 1,5$ % of the measured. value 0,3 % of the measured value 1,0 % of the measured value 0,002 m ³
Temperature at the gas volume meter: — overall uncertainty of calibration; — drift between 2 adjustments; — resolution.	$\leq \pm 2,5$ K	$\pm 1,0$ K 0,2 K 0,1 K
Absolute pressure at gas volume meter: Relative pressure at the gas volume meter; scale of the manometer: 0 Pa to –200 Pa — overall uncertainty of calibration; — resolution; — lack of fit; — drift between 2 adjustments. Atmospheric pressure maximum permissible error; —reading.	$\leq \pm 1,0$ % of the measured. value	$\pm 0,6$ Pa 0,01 Pa 1,4 % FS (full scale) 1,0 % FS ± 300 Pa 20 Pa
Absorption efficiency of the first absorber	$> 95,0$ %	98 %
Concentration q_s in the absorption solution: — standard deviation of repeatability of analysis.	$\leq 5,0$ % of the measured value	5 % of the measured value

C.5 Calculation of standard uncertainty of concentration measured

C.5.1 Model equation and application of rule of uncertainty propagation

$$C_m = \frac{q_s \times v_s \times \frac{64,1}{96,1}}{V_{m(std)}} \quad (\text{C.1})$$

with

$$V_{m(std)} = V_m \times \frac{T_{std}}{T_m} \times \frac{P_m}{P_{std}} \quad (\text{C.2})$$

where

q_s	is the mass concentration of sulphate (in milligrams per litre) collected in the sample absorption solution
v_s	is the volume of sample absorption solution (in litres)
$V_{m(std)}$	is the gas volume sampled from the gas meter, dry and at standard conditions (in cubic metres)
T_m	is the mean temperature (in Kelvin) of the sampled gas at the gas-meter
T_{std}	is the standard temperature, 273 K
$P_m = P_{ref} + P_{atm}$	is the absolute pressure (in kilopascals) at the gas volume meter; P_m is equal to the sum of relative pressure measured at the gas volume meter P_{ref} plus atmospheric pressure P_{atm}
P_{std}	is the standard pressure, 101,325 kPa
V_m	is the gas volume sampled (in cubic metre) achieved by difference between values given by the gas volume meter at the end and at the beginning of the sampling period. The value at the beginning of the sampling period corresponds to a reading of an indicator; the value at the end of the sampling period corresponds to a reading of a measured value.

Conversion of the concentration of SO₂ at O₂ reference concentration:

$$C_{m,O_2ref} = C_m \times \frac{21 - O_{2,ref}}{21 - O_{2,meas}} \quad (\text{C.3})$$

where

C_{m,O_2ref}	is the concentration at O ₂ reference concentration (mg/m ₀ ³)
C_m	is the concentration at O ₂ measured concentration in the duct (mg/m ₀ ³)
$O_{2,ref}$	is O ₂ reference concentration (in % volume)
$O_{2,meas}$	is O ₂ measured concentration in the duct (in % volume)

Calculation of the concentration:

$$V_{m(std)} = 0,049 \times \frac{273}{296,2} \times \frac{100,281}{101,325} = 0,045 \text{ m}_0^3$$

The concentration at standard conditions of temperature and pressure, and at O_2 concentration measure is equal to:

$$C_m = \frac{14,56 \times 200 \times \frac{64,1}{96,1}}{0,045} = 43,4 \text{ mg/m}_0^3$$

The concentration at standard conditions of temperature and pressure, and at O_2 reference concentration is equal to:

$$C_{m,O_2ref} = 43,4 \times \frac{21-11}{21-12,3} = 49,9 \text{ mg/m}_0^3$$

Expression for the calculation of the combined uncertainty of C_m :

$$\frac{u^2(C_m)}{(C_m)^2} = \frac{u^2(q_s)}{(q_s)^2} + \frac{u^2(v_s)}{(v_s)^2} + \frac{u^2(V_{m(std)})}{(V_{m(std)})^2} \quad (C.4)$$

Calculation of the combined uncertainty of $V_{m(std)}$:

$$V_{m(std)} = V_m \frac{T_{std}}{T_m} \frac{P_m}{P_{std}} = V_m \frac{T_{std}}{T_m} \frac{P_{ref} + P_{atm}}{P_{std}} \quad (C.5)$$

Hypothesis: we consider that uncertainties of T_{std} and P_{std} are negligible:

$$\begin{aligned} u^2(V_{m(std)}) &= \left(\frac{\partial V_{m(std)}}{\partial V_m} \right)^2 \times u^2(V_m) + \left(\frac{\partial V_{m(std)}}{\partial T_m} \right)^2 \times u^2(T_m) + \left(\frac{\partial V_{m(std)}}{\partial P_{ref}} \right)^2 \times u^2(P_{ref}) \\ &+ \left(\frac{\partial V_{m(std)}}{\partial P_{atm}} \right)^2 \times u^2(P_{atm}) \end{aligned} \quad (C.6)$$

Calculation of sensitivity coefficients:

$$\frac{\partial V_{m(std)}}{\partial V_m} = \frac{T_{std}}{T_m} \times \frac{P_{ref} + P_{atm}}{P_{std}} = \frac{V_{m(std)}}{V_m} \quad (C.7)$$

$$\frac{\partial V_{m(std)}}{\partial T_m} = V_m \cdot T_{std} \cdot \frac{P_{ref} + P_{atm}}{P_{std}} \cdot \frac{1}{T_m^2} = - \frac{V_{m(std)}}{T_m} \quad (C.8)$$

$$\frac{\partial V_{m(std)}}{\partial P_{ref}} = V_m \cdot \frac{T_{std}}{T_m} \cdot \frac{1}{P_{std}} = \frac{V_{m(std)}}{P_{ref} + P_{atm}} = \frac{V_{m(std)}}{P_m} \quad (C.9)$$

$$\frac{\partial V_{m(std)}}{\partial P_{atm}} = V_m \cdot \frac{T_{std}}{T_m} \cdot \frac{1}{P_{std}} = \frac{V_{m(std)}}{P_{ref} + P_{atm}} = \frac{V_{m(std)}}{P_m} \quad (C.10)$$

(C.2) is equivalent to:

$$u^2(V_{m(std)}) = \left(\frac{V_{m(std)}}{V_m} \right)^2 \times u^2(V_m) + \left(\frac{-V_{m(std)}}{T_m} \right)^2 \times u^2(T_m) + \left(\frac{V_{m(std)}}{P_{ref} + P_{atm}} \right)^2 \times u^2(P_{ref}) + \left(\frac{V_{m(std)}}{P_{ref} + P_{atm}} \right)^2 \times u^2(P_{atm}) \quad (C.11)$$

(C.4) is equivalent to:

$$\frac{u^2(C_m)}{(C_m)^2} = \frac{u^2(q_s)}{(q_s)^2} + \frac{u^2(v_s)}{(v_s)^2} + \frac{u^2(V_m)}{(V_m)^2} + \frac{u^2(T_m)}{(T_m)^2} + \frac{u^2(P_{ref})}{(P_{ref} + P_{atm})^2} + \frac{u^2(P_{atm})}{(P_{ref} + P_{atm})^2} \quad (C.12)$$

C.5.2 Results of the standard uncertainties calculations

Table C.4 — Results of the standard uncertainty calculations

Performance characteristic	Value of standard uncertainty at limit value)	Relative standard uncertainty
Volume of the absorption solution	$u(v_s) = \sqrt{\left(\frac{1,4}{\sqrt{3}}\right)^2 + \left(\frac{2}{2\sqrt{3}}\right)^2} + 0,5^2 = 1,112 \text{ ml}$	$\frac{u(v_s)}{v_s} = 0,006$
Determination of the concentration q_s in the absorption solution	$u(q_s) = \frac{5}{100} \times 14,56 = 0,728 \text{ mg SO}_4^{2-} / \text{l}$	$\frac{u(q_s)}{q_s} = 0,05$
Volume of sampled gas	$u^2(V_m) = u^2(cal, V_m) + u^2(rep, V_m) + u^2(dr, V_m) + 2u^2(read, V_m)$ $u(V_m) = \sqrt{\left(\frac{1,5 \times 0,049}{2 \times 100}\right)^2 + \left(\frac{0,3 \times 0,049}{100}\right)^2 + \left(\frac{1/100 \times 0,049}{\sqrt{3}}\right)^2 + 2 \times \left(\frac{0,002}{2\sqrt{3}}\right)^2} = 0,00011 \text{ m}^3$	$\frac{u(V_m)}{V_m} = 0,0210$

(continued)

Table C.4 (end)

Performance characteristic	Value of standard uncertainty at limit value)	Relative standard uncertainty
Temperature at the gas volume meter	$u^2(T_m) = u^2(cal, T_m) + u^2(res, T_m)$ $+ u^2(dr, T_m) + u^2(mean, T_m)$ $u(T_m) = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{0,1}{2\sqrt{3}}\right)^2 + \left(\frac{0,2}{\sqrt{3}}\right)^2 + 0,231^2} = 0,566 \text{ K}$	$\frac{u(T_m)}{T_m} = 0,0019$
Relative pressure at the gas meter	$u^2(P_{rel}) = u^2(cal, P_{rel}) + u^2(res, P_{rel})$ $+ u^2(lfir, P_{rel}) + u^2(dr, P_{rel}) + u^2(mean, P_{rel})$ $u(P_{rel}) = \sqrt{\left(\frac{0,6}{2}\right)^2 + \left(\frac{0,01}{2\sqrt{3}}\right)^2 + \left(\frac{1,4/100 \times 200}{\sqrt{3}}\right)^2 + \left(\frac{1/100 \times 200}{\sqrt{3}}\right)^2 + 2,03^2} = 2,027 \text{ Pa}$	$\frac{u(P_{rel})}{P_m} = 2,1 \cdot 10^{-5}$
Atmospheric pressure	$u^2(P_{atm}) = u^2(MPE, P_{atm}) + u^2(res, P_{atm})$ $u(P_{rel}) = \sqrt{\left(\frac{300}{\sqrt{3}}\right)^2 + \left(\frac{20}{2\sqrt{3}}\right)^2} = 173,3 \text{ Pa}$	$\frac{u(P_{atm})}{P_m} = 0,00173$

C.5.3 Estimation of the combined uncertainty

Uncertainty associated to C_m :

The result of the calculation of the combined uncertainty according to the equation (C.11) is:

Standard uncertainty: $u(C_m) = 2,38 \text{ mg/m}_0^3$

Overall uncertainty: $U(C_m) = \pm 4,75 \text{ mg/m}_0^3 (k = 2)$

$U(C_{m,rel}) = \pm 11,0 \% (k = 2)$

Annex D (informative)

Evaluation of the method

D.1 General

The method has been evaluated:

- during a round robin test for analytical procedures: inter-laboratory comparison of sulphate analysis of solutions;
- during 6 field tests, on waste incineration installations, co-incineration installations and large combustion plants for the method (sampling and analysis).

D.2 Round robin test of analytical methods

Twenty-two laboratories took part to the round robin test. The ion chromatography method was carried out by twenty-one participants and the titration method by six ones.

Four different samples were distributed to the participants and each solution had to be analysed three times:

- blank solution;
- “standard” solution containing $50,1 \pm 0,1 \text{ mg/l SO}_4^{2-}$;
- “absorption solution” containing $\approx 61 \text{ mg/l SO}_4^{2-}$; and
- “added solution” containing $\approx 86 \text{ mg/l SO}_4^{2-}$.

Results of standard deviation of repeatability and reproducibility obtained by the participants are given in Table D.1 and in Table D.2:

Table D.1 — Standard deviation of repeatability

Sample	Ion chromatography		Titration method	
	s_r in mg/l SO_4^{2-}	s_r in % of the measured value	s_r in mg/l SO_4^{2-}	s_r in % of the measured value
Standard solution $50,1 \text{ mg/l SO}_4^{2-}$	0,8	1,7	1,0	2,0
Absorption solution $\approx 61 \text{ mg/l SO}_4^{2-}$	1,2	2,0	0,8	1,2
Added solution $\approx 86 \text{ mg/l SO}_4^{2-}$	2,1	2,5	1,5	1,7

Table D.2 — Standard deviation of reproducibility

Sample	Ion chromatography		Titration method	
	s_R in mg/l SO_4^{2-}	s_R in % of the measured value	s_R in mg/l SO_4^{2-}	s_R in % of the measured value
Standard solution 50,1 mg/l SO_4^{2-}	2,1	4,1	2,7	5,5
Absorption solution ≈ 61 mg/l SO_4^{2-}	4,8	8,0	4,7	7,7
Added solution ≈ 86 mg/l SO_4^{2-}	5,6	6,5	6,1	7,0

D.3 Field tests

D.3.1 General

The method has been evaluated during six field tests, on waste incineration installations, co-incineration installations and large combustion plants. Each test was performed by at least four different European measuring teams originating from ten member countries.

D.3.2 Characteristics of installations

- 1st field test: INERIS bench-loop at Verneuil en Halatte (F); the bench-loop simulates combustion or waste incineration exhaust gases. Five teams took part in the 1st field test. Double measurements were not performed simultaneously but sequentially. Five different flue gas matrices were generated. Within each matrix, two sequential measurements were performed. Two additional sequential measurements were performed in flue gas matrices where the flue gas concentrations varied. There was a total of twelve measurements performed by all the teams;
- 2nd field test: waste incinerator in Denmark. Four teams took part to the field test and performed double measurements simultaneously. A total of sixteen measurements was performed by all the teams;
- 3rd field test: waste incinerator in Italy. Four teams took part to the field test. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements was performed by each pair of two teams while a total of twelve single measurements was performed by all teams;
- 4th field test: co-incinerator combined heat and power installation in Sweden. The fluidised bed boilers operate on fuel mixes of wood chips, demolition waste, peat and coal. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements was performed by each pair of two teams while a total of twelve single measurements was performed by all the teams;
- 5th field test: co-incinerator cement plant in Germany. The fuel could be coal, heavy oil and secondary fuel (e.g. paper, plastics, textiles, and tires). Four teams took part to the field test and performed double measurements simultaneously. A total of sixteen double measurements was performed by all the teams;
- 6th field test: coal fired power plant in Germany. Four teams performed their double measurements simultaneously. The total amount of double measurements performed by all teams was 12.

An overview of the flue gas characteristics is given in Table D.3.

Table D.3 — Flue gas characteristics during field tests

Field test	Installation	Fuel	Flue gas characteristics						
			T °C	O ₂ % vol	NO _x mg/m ³	SO ₂ mg/m ³	CO mg/m ³	H ₂ O % vol	PM mg/m ³
1	Power plant ^a	Natural gas	< 150	5 to 13	10 to 1 300	10 to 2 000	20 to 400	10 to 21	<1
2	Waste incinerator	Municipal waste	90 to 110	8 to 11	180 to 250	25 to 250	5 to 15	13 to 19	1 to 5
3	Waste incinerator	Municipal waste	85 to 105	16 to 18	61 to 78	5 to 50	0 to 2	8 to 12	1 to 5
4	Co-incinerator	Wood, waste, peat, coal	70 to 80	4 to 6	4 to 70	0 to 10	50 to 150	8 to 12	0 to 20
5	Co-incinerator	Coal, oil, waste	140 to 170	4 to 6	440 to 1 060	60 to 170	260 to 740	23 to 26	5 to 10
6	Power plant	Coal	130 to 140	8,9 to 9,2	110 to 140	1000 to 1130	3 to 6	5,5 to 8	< 50
^a Bench-loop: flue gas simulation.									

D.3.3 Limits of detection

During the field tests the following detection limits have been given:

- ion chromatography analyse: 0,05 mg SO₄²⁻/l to 0,2 mg SO₄²⁻/l, with an average of 0,1 mg SO₄²⁻/l;
- titration method: 1 mg SO₄²⁻/l to 10 mg SO₄²⁻/l. It seems that with a well trained personal 1 mg SO₄²⁻/l can be reached.

D.3.3.1 Repeatability and reproducibility

D.3.3.2 General

Repeatability standard deviation s_r and reproducibility standard deviation s_R are determined by performing inter-laboratory tests.

Repeatability standard deviation s_r , repeatability confidence interval (CI_r) and repeatability in the field r are calculated according to ISO 5725-2 and ISO 5725-6, from the results of the double measurements implemented by the same laboratory.

$$CI_r = t_{0,95; n-1} \times s_r \quad \text{and} \quad r = \sqrt{2} \times t_{0,95; n-1} \times s_r$$

where

CI_r is the repeatability confidence interval

s_r is the repeatability standard deviation

$t_{0,95; n-1}$ is the student factor for a level of confidence of 95 % and a degree of freedom of $n - 1$ (n : number of double measurements)

r is the repeatability in the field

Reproducibility standard deviation s_R , reproducibility confidence interval (CI_R) and reproducibility in the field R are calculated according to ISO 5725-2, from the results of parallel measurements performed simultaneously by several laboratories.

$$CI_R = t_{0,95; np-1} \times s_R \quad \text{and} \quad R = \sqrt{2} \times t_{0,95; np-1} \times s_R$$

where

CI_R is the external confidence interval

s_R is the reproducibility standard deviation

$t_{0,95; np-1}$ is the student factor for a level of confidence of 95 % and a degree of freedom of $np-1$ (n : number of measurements; p : number of laboratories)

R is the reproducibility in the field

D.3.3.3 Repeatability

Table D.4 — Repeatability in the field for ion chromatography method

Field test	Concentration		Number of teams	Number of double measurements	Repeatability standard deviation s_r	95 % confidence interval for a single measurement CI_R			Repeatability r	
	Range mg/m ³	Average mg/m ³			mg/m ³	mg/m ³	%	mg/m ³	%	
1A	16 to 27	21	5	2	2,9	6,6	31,4	9,3	45	
1B	40 to 63	54	5	2	3,1	7,0	13,0	9,9	18	
1C	200 to 241	219	5	2	4,8	11	5,0	15,4	7	
1D	884 to 1 038	961	5	2	26	58	6,2	82,4	9	
1E	1 889 to 2 174	2 088	5	2	100	226	10,8	319	15	
1F	871 to 1 033	930	5	1	24	68	7,3	95,8	10	
1G	180 to 221	198	5	1	4,5	12	6,1	17,7	9	
2	25 to 210	87	4	16	0,029 [SO ₂] + 1,18	8,2	9,4	11,6	13	
3	6 to 50	27	4	12	0,076 [SO ₂] + 0,11	4,7	17,4	6,6	25	
4	0,4 to 9	3,5	2	5	0,49	1,4	40	2,0	57	
5	64 to 169	108	4	12	7,5	16	15,3	22,6	21	
6	1 000 to 1 123	1 055	4	12	37	82	7,8	116	11	

The following functions were determined:

$$s_r(C) = 0,0338 C + 1,52 \quad \text{in mg/m}^3$$

$$s_{r \text{ limit}}(C) = 0,051 C + 2,3 \quad \text{in mg/m}^3$$

$$CI_r(C) = 0,08 C + 3 \quad \text{in mg/m}^3$$

$$r(C) = 0,113 C + 4,2 \quad \text{in mg/m}^3$$

Table D.5 — Repeatability in the field for Thorin method

Field test	Concentration		Number of teams	Number of double measurements	Repeatability standard deviation s_r	95 % confidence interval for a single measurement CI_r		Repeatability r	
	Range mg/m ³	Average mg/m ³			mg/m ³	mg/m ³	%	mg/m ³	%
1A	17 to 27	22	4	2	0,34	0,8	3,6	1,1	5,1
1B	40 to 63	55	4	2	1,2	2,8	5,1	3,9	7,1
1C	207 to 241	220	4	2	5,3	12,5	5,7	17,7	8,0
1D	869 to 1 031	946	4	2	23	55,4	5,9	78,4	8,3
1E	1 889 to 2 299	2 098	4	2	52	123	5,9	175	8,3
1F	795 to 1 025	926	4	1	23	73	7,9	103	11,1
1G	191 to 221	209	4	1	5	15,9	7,6	22,5	10,8
2	26 to 204	86	4	16	0,034 [SO ₂] + 1,15	8,7	10,1	12,3	14,3
3	9 to 51	30	4	12	0,053 [SO ₂] + 1,8	7,5	25	10,6	35,4
4	2,8 to 9,6	5,6	2	11	1,6	3,6	64,3	5,0	90,9
5	70 to 170	111	3	12	6,8	15	13,5	21,2	19,1
6	973 to 1 089	1 018	4	12	30	66	6,5	93,3	9,2

The following functions were determined:

$$s_r(C) = 0,0273 C + 2,4 \quad \text{in mg/m}^3$$

$$s_{r \text{ limit}}(C) = 0,051 C + 2,3 \quad \text{in mg/m}^3$$

$$CI_r(C) = 0,06 C + 3 \quad \text{in mg/m}^3$$

$$r(C) = 0,085 C + 4,2 \quad \text{in mg/m}^3$$

D.3.3.4 Reproducibility

Table D.6 — Reproducibility in the field for ion chromatography method

Field test	Concentration		Number of teams	Number of double measurements	Reproducibility standard deviation s_R	95 % confidence interval for a single measurement CI_R		Reproducibility R	
	Range mg/m ³	Average mg/m ³			mg/m ³	mg/m ³	%	mg/m ³	
1A	16 to 27	21	5	2	4,0	9,0	42,9	13	62
1B	40 to 63	54	5	2	7,0	16	29,7	22	41
1C	200 to 241	219	5	2	15	33	15,1	46	21
1D	884 to 1 038	961	5	2	74	167	17,4	236	25
1E	1 889 to 2 174	2 088	5	2	137	311	14,9	440	21
1F	871 to 1 033	930	5	1	102	282	30,3	398	43
1G	180 to 221	198	5	1	17	47	23,8	66	33
2	25 to 210	87	4	16	0,044 [SO ₂] + 3,5	15,4	17,7	22	25
3	6 to 50	27	4	12	0,032 [SO ₂] + 1,8	5,9	21,8	8,3	31
4	0,4 to 9	3,5	4	12	0,78	1,7	48,6	2,4	69
5	64 to 169	108	3	12	8,9	20	18,5	28	26
6	1 000 to 1 123	1 055	4	12	56	123	11,7	174	16

The following functions were determined:

$$s_R(C) = 0,051\,4\,C + 2 \quad \text{in mg/m}^3$$

$$CI_R(C) = 0,113\,C + 5,4 \quad \text{in mg/m}^3$$

$$R(C) = 0,16\,C + 7,6 \quad \text{in mg/m}^3$$

Table D.7 — Reproducibility in the field for Thorin method

Field test	Concentration		Number of teams	Number of double measurements	Reproducibility standard deviation s_R	95 % confidence interval for a single measurement CI_R		Reproducibility R	
	Range mg/m ³	Average mg/m ³			mg/m ³	mg/m ³	%	mg/m ³	
1A	17 to 27	22	4	2	4,0	9,4	42,7	13	61
1B	40 to 63	55	4	2	8,1	19	34,5	27	49
1C	207 to 241	220	4	2	15	35	16,0	49	22
1D	869 to 1 031	946	4	2	76	179	18,9	253	27
1E	1 889 to 2 299	2 098	4	2	181	427	20,4	604	29
1F	795 to 1 025	926	4	1	112	357	38,6	504	54
1G	191 to 221	209	4	1	5,0	16	7,7	22	11
2	26 to 204	86	4	16	0.088 [SO ₂] + 2,37	20,8	24,2	30	35
3	9 to 51	30	4	12	-0.004 9 [SO ₂] + 3,6	8	26,7	11	36
4	2,8 to 9,6	5,6	4	8	2,2	5,1	91	7	129
5	70 to 170	111	3	12	8,3	18	16,2	25	23
6	973 to 1 089	1 018	4	12	47	103	10,1	147	14

The following functions were determined:

$$s_R(C) = 0,043 C + 3,4 \quad \text{in mg/m}^3$$

$$CI_R(C) = 0,089 C + 6 \quad \text{in mg/m}^3$$

$$R(C) = 0,126 C + 8,5 \quad \text{in mg/m}^3$$

D.3.4 Absorption efficiency

Absorption efficiency has been determined for concentration in the range 20 mg/m³ to 2000 mg/m³ SO₂.

Sampling conditions was:

- volume of sampled gas during one half hour: 45 l to 200 l;
- volume of absorption solution in each absorber: 25 ml to 80 ml;
- all results were better than the requirement of 95 %.

Annex E
(informative)

Test of equivalency of Thorin method and ion chromatography

In this example, we provide data to check repeatability and trueness of Thorin method compared to those of the Ion chromatography method, considered as the reference method. This table was set up according to the rules given in CEN TS 14793.

Concentrations are expressed in mg/m³.

Table E.1— Ion Chromatography

Test [I]	Parallel measurements		Nb/meas.	Means		Variances	$(Z_{i1} - \bar{Z}_i)^2$	$(Z_{i1} - \bar{\bar{Z}})^2$	$(Z_{i2} - \bar{\bar{Z}})^2$	$(Z_{i1} - \bar{\bar{Z}})^2$	$(Z_{i2} - \bar{\bar{Z}})^2$
	Z_{i1}	Z_{i2}		$\bar{Z}_i$	$\bar{Z}_i^2$						
1	43,81	46,32	2	45,1	203 0,9	3,150 05	1,58	-12,13	-9,62	147,11	92,52
2	33,84	36,26	2	35,1	122 8,5	2,928 2	1,46	-22,10	-19,68	488,36	387,26
3	97,30	99,32	2	98,3	966 4,9	2,040 2	1,02	41,36	43,38	1 710,75	1 881,93
4	203,72	211,26	2	207,5	43 052,1	28,425 8	14,21	147,78	155,32	21 839,28	24 124,68
5	57,07	58,64	2	57,9	3 347,2	1,232 45	0,62	1,13	2,70	1,28	7,30
6	156,67	176,66	2	171,2	29 297,5	60,390 05	30,20	109,79	120,72	12 040,94	14 573,61
7	42,40	43,85	2	43,1	1 859,8	1,051 25	0,53	-13,54	-12,09	183,30	146,14
8	89,26	88,62	2	88,9	7 910,3	0,204 8	0,10	33,32	32,68	1 110,30	1 068,06
9	87,16	86,56	2	86,9	7 544,7	0,18	0,09	31,22	30,62	974,76	937,66
10	51,93	55,01	2	53,5	2 859,0	4,743 2	2,37	-4,01	-0,93	16,07	0,86
11	22,37	22,59	2	22,5	505,4	0,024 2	-33,57	-33,35	1 126,86	1 126,86	1 112,14
(continued)											

Table E.1 — Ion Chromatography (end)

Test [I]	Parallel measurements		Nb/meas.	Means		Variances	$(Z_{i1} - \bar{Z}_i)^2$	$(Z_{i1} - \bar{\bar{Z}})$	$(Z_{i2} - \bar{\bar{Z}})$	$(Z_{i1} - \bar{\bar{Z}})^2$	$(Z_{i2} - \bar{\bar{Z}})^2$
	Z_{i1}	Z_{i2}		$\bar{Z}_i$	$\bar{Z}_i^2$						
			n_i			$S_i^2(Z)$					
12	108,73	108,82	2	108,8	11 832,0	0,004 05	0,00	52,79	52,88	2 786,91	2 796,42
13	133,55	139,72	2	136,6	18 669,1	19,034 45	9,52	77,61	83,78	6 023,50	7 019,29
14	51,65	55,71	2	53,7	2 881,5	8,241 8	4,72	-4,29	-0,23	18,39	0,05
15	143,31	146,64	2	145,0	21 017,8	5,544 45	2,77	87,37	90,70	7 633,73	8 226,71
16	4,20	3,80	2	4,0	16,0	0,08	0,04	-51,74	-52,14	2 676,90	2 718,45
17	3,50	3,20	2	3,4	11,2	0,045	0,02	-52,44	-52,74	1 749,83	2 781,38
18	2,00	1,80	2	1,9	3,6	0,02	0,01	-53,94	-54,14	2 909,39	2 931,01
19	1,40	1,30	2	1,4	1,8	0,005	0,00	-54,54	-54,64	2 974,48	2 985,40
20	1,30	1,20	2	1,3	1,6	0,005	0,00	-54,64	-54,76	2 985,40	2 996,34
21	5,80	5,30	2	5,6	0,125	0,06	0,06	-50,14	-50,64	2 513,90	2 564,29
22	6,80	6,30	2	6,6	42,9	0,125	0,06	-49,14	-49,64	2 414,62	2 464,01
23	8,70	8,00	2	8,4	69,7	0,245	0,12	-47,24	-47,94	2 231,50	2 298,13
24	10,00	9,30	2	9,7	93,1	0,245	0,12	-45,94	-46,64	2 110,37	2 175,18
25	3,79	1,50	2	2,6	7,0	2,622 05	1,31	-52,44	-54,44	2 719,50	2 963,58
$N = 50$				55,94	85 749,6	3 572,90	5,63				
				$\bar{\bar{Z}}$	$SSD(\bar{\bar{Z}})$	$S_2(\bar{\bar{Z}})$	$S_r^2(Z)$				

Table E.2 — Thorin method

Test	Parallel measurements		Nb/meas.	Means	Variances		$(X_{i1} - \bar{X}_i)^2$	$(X_{i1} - \bar{\bar{X}})$	$(X_{i2} - \bar{\bar{X}})$	$(X_i - \bar{\bar{X}})^2$	$(X_{i2} - \bar{\bar{X}})^2$
	X_{i1}	X_{i2}			$S_i^2(X)$	$\bar{X}_i - \bar{\bar{X}}_i$					
			n_i	$\bar{X}_i$	$\bar{\bar{X}}_i^2$	$\bar{X}_i - \bar{\bar{X}}_i$					$(X_{i2} - \bar{\bar{X}})^2$
1	45,54	49,64	2	4,6	2 264,8	2 144,643 35	4,2025	-12,80	-8,70	163,74	75,62
2	36,9	38,61	2	37,3	1 394,6	1 308,942 25	1,600 225	-22,26	-19,73	495,33	389,12
3	99,56	101,36	2	100,5	10 092,2	9 876,222 6	0,81	41,22	43,02	1 699,42	1 851,06
4	205,57	213,09	2	209,3	43 819,0	43 433,881 7	14,137 6	147,23	154,75	21 677,85	23 948,80
5	59,66	60,45	2	60,1	3 606,6	3 474,482 025	0,156 025	1,32	2,11	1,75	4,47
6	173,04	180,16	2	176,6	31 187,6	20 227,739	12,673 6	114,70	121,82	13 157,01	14 841,09
7	45,77	47,77	2	46,8	2 187,4	2 016,956 25	1	-12,57	-10,57	157,90	111,64
8	93,63	95,25	2	94,4	8 918,9	8 399,493 6	0,656 1	35,29	36,91	1 245,67	1 362,64
9	88,85	89,09	2	89,0	7 915,7	7 727,934 2	0,014 4	30,51	30,75	931,10	945,81
10	55,47	58,38	2	56,9	32,40,5	3 043,779 75	2,117 025	-2,87	0,04	8,21	0,00
11	24,70	26,44	2	25,6	653,8	574,813 6	0,756 9	-33,64	-31,90	1 131,38	1 017,35
12	109,29	112,21	2	110,8	12 265,6	12 046,831 25	2,13 16	50,95	53,87	2 596,31	2 902,41
13	136,94	150,22	2	143,6	20 615,2	19 618,053 3	44,089 6	78,60	91,88	6 178,59	8 442,67
14	54,84	59,15	2	57,0	3 248,4	3 059,491 6	4,644 025	-3,50	0,81	12,22	0,66
15	114,65	147,19	2	130,9	17 140,0	18 980,127	264,712 9	56,31	88,85	3 174,27	7 895,03
16	6,70	6,10	2	6,4	41,0	25,6	0,09	-51,64	-52,24	2 666,28	2 728,60
17	5,00	4,60	2	4,8	23,0	16,08	0,04	-53,34	-53,74	2 844,73	2 887,56
18	4,30	3,90	2	4,1	16,8	7,79	0,04	-54,04	-54,44	2 919,89	2 963,28
19	4,20	3,90	2	4,1	16,4	5,4675	0,022 5	-54,14	-54,44	2 930,71	2 963,28
20	4,20	3,90	2	4,1	16,4	5,0625	0,022 5	-54,14	-54,44	2 930,71	2 963,28

(continued)

Table E.2 — Thorin method (end)

Test	Parallel measurements		Nb/meas.	Means		Variances						
[i]			n_i	$\bar{X}_i$	$\bar{\bar{X}}_i^2$	$S_i^2(X)$	$\bar{X}_i - \bar{Z}_i$	$(X_{i1} - \bar{X}_i)^2$	$(X_{i1} - \bar{\bar{X}})$	$(X_{i2} - \bar{\bar{X}})$	$(X_i - \bar{\bar{X}})^2$	
	X_{i1}	X_{i2}										$(X_{i2} - \bar{\bar{X}})^2$
21	9,80	9,10	2	9,5	89,3	0,245	52,447 5	0,122 5	-48,54	-49,24	2 355,74	2 424,18
22	8,70	8,10	2	8,4	70,6	0,18	55,02	0,09	-49,64	-50,24	2 463,73	2 523,66
23	11,20	10,40	2	10,8	116,6	0,32	90,18	0,16	-47,14	-47,94	2 221,80	2 297,86
24	11,10	10,40	2	10,8	115,6	0,245	103,737 5	0,122 5	-47,24	-47,94	2 231,24	2 297,86
25	9,60	9,00	2	9,3	86,5	0,18	24,598 5	0,09	-48,74	-49,34	2 375,20	2 434,04
	$N = 50$			58,34	84 065,4	3 502,72	84 738,23	28,36	0,998 055 9	0,990 131	6 753,59	
				$\bar{\bar{X}}$	$SSD(X_b)$	$S_2(\bar{X})$	$SPD(\bar{B}i\bar{Z})$	$S_r^2(X)$	R	C_1	$S_r^2(X)$	

$X_{15,1}$ seems to be an outlier – a Grubbs test has to be applied

Table E.3 — Detection of outliers

X_{i1}	X_{i2}	$Dx_i = X_{i1} - X_{i2}$	$Dx_i / x_{iav} \times 100$	G_i
45,54	49,64	-4,10	-8,62	-0,99
36,08	38,61	-2,53	-6,77	-0,77
99,56	101,36	-1,80	-1,79	-0,16
205,57	213,09	-7,52	-3,59	-0,38
59,66	60,45	-0,79	-1,32	-0,10
173,04	180,16	-7,12	-4,03	-0,44
45,77	47,77	-2,00	-4,28	-0,47
93,63	95,25	1,62	-1,72	-0,15
88,85	89,09	-0,24	-0,27	0,02
55,47	58,38	-2,91	-5,11	-0,57
24,70	26,44	-1,74	-6,80	-0,77
109,29	112,21	-2,92	-2,64	-0,27
136,94	150,22	-13,28	-9,25	-1,07
54,84	59,15	-4,31	-7,56	-0,87
114,65	147,19	-32,54	-24,85	-2,97
6,70	6,10	0,60	9,38	1,20
5,00	4,60	0,40	8,33	1,07
4,30	3,90	0,40	9,76	1,24
4,20	3,90	0,30	7,41	0,96
4,20	3,90	0,30	7,41	0,96
9,80	9,10	0,70	7,41	0,96
8,70	8,10	0,60	7,14	0,93
11,20	10,40	0,80	7,41	0,96
11,10	10,40	0,70	6,51	0,85
9,60	9,00	0,60	6,45	0,84

-0,46 X_{av}

8,21 s

MAX (G_i) ⇒	G 15 The value x =	2,97 15,1 can be skipped	> 2,822 Grubbs at 95 %
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Concentrations are expressed in mg/m³

Table E.4 — Ion Chromatography

Test no	Parallel measurements		Nb/meas.	Means		Variances	$(Z_{i1} - \bar{Z}_i)^2$	$(Z_{i1} - \bar{\bar{Z}})^2$	$(Z_{i2} - \bar{\bar{Z}})^2$	$(Z_{i2} - \bar{\bar{Z}})^2$
	Z_{i1}	Z_{i2}		$\bar{Z}_i$	$\bar{Z}_i^2$					
			n_i			$S_i^2(Z)$				
1	43,81	46,32	2	45,1	2 030,9	3,150 05	1,58	-8,42	-5,91	34,92
2	33,84	36,26	2	35,1	1 228,5	2,928 2	1,46	-18,39	-15,97	255,01
3	97,30	99,32	2	98,3	9 664,9	2,040 2	1,02	45,07	47,09	2 217,57
4	203,72	211,26	2	207,5	43 052,1	28,425 8	14,21	151,49	159,03	25 290,87
5	57,07	58,64	2	57,9	3 347,2	1,234 5	0,62	4,84	6,41	41,10
6	165,67	176,66	2	171,2	29 297,5	60,390 05	30,20	113,44	124,43	15 483,08
7	42,40	43,85	2	43,1	1 859,8	1,051 25	0,53	-9,83	-8,38	70,21
8	89,26	88,62	2	88,9	7 910,3	0,204 8	0,10	37,03	36,39	1 324,31
9	87,16	86,56	2	86,9	7 544,7	0,18	0,09	34,93	34,33	1 178,62
10	51,93	55,01	2	53,5	2 859,0	4,743 2	2,37	-0,30	2,78	7,73
11	22,37	22,59	2	22,5	505,4	0,024 2	0,01	-29,86	-29,64	878,47
12	108,73	108,82	2	108,8	11 832,0	0,004 05	0,00	56,50	56,59	3 202,55
13	133,55	139,72	2	136,6	18 669,1	19,034 45	9,52	81,32	87,49	7 654,68
14	51,65	55,71	2	53,7	2 881,5	8,241 8	4,12	-0,58	3,48	12,12
16	4,20	3,80	2	4,0	16,0	0,08	0,04	-48,03	-48,43	2 345,36
17	3,50	3,20	2	3,4	11,2	0,045	0,02	-48,73	-49,03	2 403,84
18	2,00	1,80	2	1,9	3,6	0,02	0,01	-50,23	-50,43	2 543,08
19	1,40	1,30	2	1,4	1,8	0,005	0,00	-50,83	-50,93	2 593,76
20	1,30	1,20	2	1,3	1,6	0,005	0,00	-50,93	-51,03	2 603,95

(continued)

Table E.4 — Ion Chromatography (end)

Test no	Parallel measurements		Nb/meas.	Means		Variances	$(Z_{i1} - \bar{Z}_i)^2$	$(Z_{i1} - \bar{\bar{Z}})^2$	$(Z_{i2} - \bar{\bar{Z}})^2$	$(Z_{i1} - \bar{\bar{Z}})^2$	$(Z_{i2} - \bar{\bar{Z}})^2$
[I]	Z_{i1}	Z_{i2}	n_i	$\bar{Z}_i$	$\bar{Z}_i^2$	$S_i^2(Z)$	$(Z_{i1} - \bar{Z}_i)^2$				
21	5,80	5,30	2	5,6	30,8	0,125	0,06	-46,43	-46,93	2 155,65	2 202,33
22	6,80	6,30	2	6,6	42,9	0,125	0,06	-45,43	-45,93	2 063,79	2 109,47
23	8,70	8,00	2	8,4	69,7	0,245	0,12	-43,53	-44,23	1 894,77	1 956,20
24	10,00	9,30	2	9,7	93,1	0,245	0,12	-42,23	-42,93	1 783,26	1 842,90
25	3,79	1,50	2	2,6	7,0	2,622 05	1,31	-48,44	-50,73	2 346,33	2 573,43
$N = 48$				52,23	77 491,8	3 369,21	5,63				
				$\bar{\bar{Z}}$	$SSD(\bar{\bar{Z}})$	$S_2(\bar{\bar{Z}})$	$S_r^2(Z)$				

Table E.5 — Thorin method

Test no [i]	Parallel measurements		Nb/meas.	Means	Variances		$\bar{X}_i - \bar{Z}_i$	$(X_{i1} - \bar{X}_i)^2$	$(X_{i1} - \bar{\bar{X}})^2$	$(X_{i2} - \bar{\bar{X}})$	$(X_i - \bar{\bar{X}})^2$	$(X_{i2} - \bar{\bar{X}})^2$
	X_{i1}	X_{i2}			$\bar{X}_i$	$\bar{\bar{X}}_i^2$		$S_i^2(X)$				
1	45,54	49,64	2	47,60	2 264,8	2 264,8	2 144,643 35	8,405	4,202 5	-9,77	95,49	32,17
2	36,08	38,61	2	37,3	1 394,6	1 394,6	1 308,942 25	3,200 45	1,600 225	-19,23	369,86	278,95
3	99,56	101,36	2	100,5	10 092,2	10 092,2	9 876,222 6	1,62	0,81	44,25	1 957,92	2 120,45
4	205,57	213,09	2	209,3	43 819,0	43 819,0	43 433,881 7	28,275 2	14,137 6	150,26	22 577,57	24 894,00
5	59,66	60,45	2	60,1	3 606,6	3 606,6	3 474,482 03	0,312 05	0,156 025	4,35	18,91	26,40
6	173,04	180,16	2	176,6	31 187,6	31 187,6	30 227,739	25,347 2	12,673 6	117,73	13 859,96	15 587,11
7	45,77	47,77	2	46,8	2 187,4	2 187,4	2 016,956 25	2	1	-9,54	91,04	56,88
8	93,63	95,25	2	94,4	8 918,9	8 918,9	8 399,493 6	1,312 2	0,656 1	38,32	1 468,29	1 595,07
9	88,85	89,85	2	89,0	7 915,7	7 915,7	7 727,934 2	0,028 8	0,014 4	33,54	1 124,82	1 140,98
10	55,47	58,38	2	56,9	3 240,5	3 240,5	3 043,779 75	4,234 05	2,117 025	0,16	0,03	9,41
11	24,70	26,44	2	25,6	653,8	653,8	574,813 6	1,513 8	0,756 9	-30,61	937,07	833,57
12	109,29	112,21	2	110,8	12 265,6	12 265,6	12 046,831 3	4,263 2	2,131 6	53,98	2 913,66	3 237,42
13	136,94	150,22	2	143,6	20 615,2	20 615,2	19 618,053 3	88,179 2	44,089 6	91,63	6 663,18	9 007,59
14	54,84	59,15	2	57,0	3 248,4	3 248,4	3 059,491 6	9,288 05	4,644 025	-0,47	0,22	14,73
16	6,70	6,10	2	6,4	41,0	41,0	25,6	0,18	0,09	-48,61	2 363,09	2 421,79
17	5,00	4,60	2	4,8	23,0	23,0	16,08	0,08	0,04	-50,31	2 531,26	2 571,67
18	4,30	3,90	2	4,1	16,8	16,8	7,79	0,08	0,04	-51,01	2 602,19	2 643,16
19	4,20	3,90	2	4,1	16,4	16,4	5,467 5	0,045	0,022 5	-51,11	2 612,40	2 643,16
20	4,20	3,90	2	4,1	16,4	16,4	5,062 5	0,045	0,022 5	-51,11	2 612,40	2 643,16
21	9,80	9,10	2	9,5	89,3	89,3	52,447 5	0,245	0,122 5	-45,51	2 071,31	2 135,52

(continued)

Table E.5 — Thorin method (end)

Test no												
[i]	Parallel measurements		Nb/meas.	Means		Variances						
	X_{i1}	X_{i2}	n_i	$\bar{X}_i$	$\bar{\bar{X}}_i^2$	$S_i^2(X)$	$\bar{X}_i - \bar{Z}_i$	$(X_{i1} - \bar{X}_i)^2$	$(X_{i1} - \bar{\bar{X}})$	$(X_{i2} - \bar{\bar{X}})$	$(X_i - \bar{\bar{X}})^2$	$(X_{i2} - \bar{\bar{X}})^2$
22	8,70	8,10	2	8,4	70,6	0,18	55,02	0,09	-46,61	-47,21	2 172,65	2 228,94
23	11,20	10,40	2	10,8	116,6	0,32	90,18	0,16	-44,11	-44,91	1 945,84	2 017,06
24	11,10	10,40	2	10,8	115,6	0,245	103,737 5	0,1225	-44,21	-44,91	1 954,67	2 017,06
25	9,60	9,00	2	9,3	86,5	0,18	24,598 5	0,09	-45,71	-46,31	2 089,56	2 144,77
			$N = 48$	55,31	78 577,4	3 416,41	78 006,35	7,48	0,999 662 0	1,006 980	6 555,60	
				$\bar{\bar{X}}$	$SSD(X_b)$	$S_2(\bar{X})$	$SPD(\bar{B}i\bar{Z})$	$S_r^2(X)$	R	C_1	$S_r^2(X)$	

Conclusion on the final tests of equivalence between the Thorin method and the ion chromatographic method

$$C_1 = 1,006980$$

$$C_0 = -4,9 \text{ mg/m}^3$$

Test of repeatability

$$S_r(Z)_{limit} = 0,051 Z + 2,3$$

$$\bar{\bar{Z}} = 52,23 \text{ mg/m}^3 \Rightarrow S_r(\bar{\bar{Z}})_{limit} = 5,15 \text{ mg/m}^3$$

$$S_r(Z)_{limit} = 9,50 \%$$

$$S_r(X) = 5,23 \% \quad S_r(X) < S_r(Z)_{limit}$$

$$S_r(Z) = 4,54 \% \quad S_r(Z) < S_r(Z)_{limit}$$

AM repeatability is accepted

Test of systematic deviation

$$S_r(Z) = 0,0514 z + 1,99$$

$$\bar{\bar{Z}} = 52,23 \text{ mg/m}^3 \Rightarrow S_R(\bar{\bar{Z}}) = 4,67 \text{ mg/m}^3$$

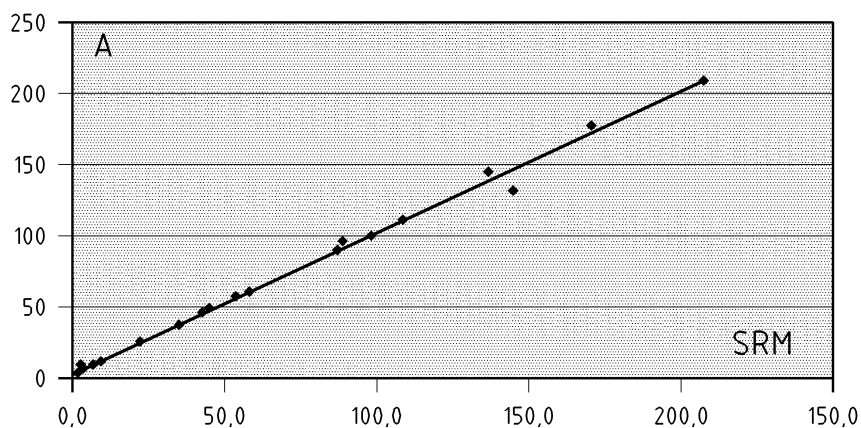
$$\Rightarrow C_0 < S_R(\bar{\bar{Z}})$$

$$r = 0,9997 > 0,97$$

$$1 - S_R(\bar{\bar{Z}})/\bar{\bar{Z}} = 0,91 \quad \text{and} \quad 1 + S_R(\bar{\bar{Z}}) = 1,09$$

$$C_1 = 1,006 \quad \Rightarrow 1 - S_R(\bar{\bar{Z}})/\bar{\bar{Z}} < C_1 < 1 + S_R(\bar{\bar{Z}})/\bar{\bar{Z}}$$

The Thorin method shows no systematic deviation compared to the Ion chromatography method.



Annex ZA (informative)

Relationship with EU Directives

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association and supports essential requirements of the EU Council Directive 2000/76/EC on the Incineration of Hazardous Waste and of the EU Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants.

WARNING — Other requirements and EU Directives may be applicable falling within the scope of use of this European Standard.

Bibliography

- [1] VIM:1994, *International vocabulary of basic and general terms in metrology*.
- [2] ISO 5725-2:1994, *Accuracy (trueness and precision) of measurement methods and results — Part 2: Basic method for the determination of the trueness of a standard measurement method*.
- [3] ISO 5725-6:1994, *Accuracy (trueness and precision) of measurement methods and results — Part 6: Use in practice of accuracy values*.
- [4] Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants.
- [5] Council Directive 2000/76/EC on waste incineration plants.
- [6] EN ISO 14956, *Air Quality — Evaluation of the suitability of a measurement procedure by comparison with a required measurement uncertainty (ISO 14956:2002)*.

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